The Cepheid Bias: Resolving the Hubble Tension

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Abstract

The Hubble Tension—the persistent 5σ discrepancy between local distance-ladder measurements ($H_0 \approx 73$ km/s/Mpc) and early-universe CMB inference ($H_0 = 67.4 \pm 0.5$ km/s/Mpc)—represents a significant challenge in precision cosmology. This paper tests whether a component of the Hubble tension can be represented as an environment-dependent Cepheid clock bias, as predicted by the Temporal Equivalence Principle (TEP).

This study tests the hypothesis that the discrepancy arises from a violation of the isochrony axiom—the assumption that proper time accumulation is independent of the local gravitational environment. Under scalar-tensor theories that break the Strong Equivalence Principle (such as TEP), Cepheid variable stars act as environment-dependent "standard clocks." In deep gravitational potentials (high velocity dispersion σ) and active-shear environments, enhanced scalar field activity is predicted to induce period contraction relative to calibration environments. When interpreted through a universal Period-Luminosity relation, this clock-rate anomaly would mimic diminished luminosity, leading to underestimated distances and an inflated local Hubble constant.

Analysis of the SH0ES Cepheid sample ($N = 29$), stratified by host galaxy velocity dispersion (a TEP-independent kinematic observable), reveals a correlation between host potential depth and derived H_0 (Spearman $\rho = 0.517$, $p = 0.0041$; Pearson $r = 0.466$, $p = 0.0109$). A median-split stratification at $\sigma_{\text{med}} \approx 96$ km/s yields $H_0 = 66.26 \pm 2.10$ km/s/Mpc (low-σ; $N = 15$) versus 74.12 ± 1.30 km/s/Mpc (high-σ; $N = 14$), implying $\Delta H_0 = 7.86$ km/s/Mpc. Because published σ values are heterogeneous (direct stellar absorption and calibrated HI linewidth proxies), all values are fully traceable to original literature sources with ADS bibcodes. Measurement methodology is treated as a first-class provenance variable and covariance-aware significance tests are reported using the full SH0ES GLS distance-modulus covariance. Significance is confirmed using non-parametric Monte Carlo permutations that fully propagate the SH0ES GLS distance-modulus covariance, yielding a robust environmental correlation ($p_{\text{cov}} \approx 0.0045$ Spearman; $p_{\text{cov}} \approx 0.023$ Pearson).

Application of the TEP conformal correction $\Delta\mu = \kappa_{\text{Cep}} \cdot S(\rho) \cdot (\sigma^2 - \sigma^2_{\text{ref}})/c^2$—derived from the TEP period-contraction combined with the virial relation $|\Phi| \propto \sigma^2$—with Observable Response Coefficient $\kappa_{\text{Cep}} = (0.99 \pm 0.56) \times 10^6$ mag and *effective calibrator reference $\sigma_{\text{ref}} = 75.25$ km/s yields a unified local Hubble constant. Out-of-sample validation (leave-one-out cross-validation, LOOCV) predicts $H_0^{\text{LOOCV}} = 67.95 \pm 1.32$ km/s/Mpc, corresponding to a Planck tension of 0.39σ; this is the primary non-circular validation because the response coefficient is trained on 28 hosts and tested on the held-out host. The in-sample corrected mean is $H_0 = 68.13$ km/s/Mpc (bootstrap mean 68.06 ± 1.49, Planck tension 0.47σ), with the corrected $r \simeq 0$ a fitted-correction diagnostic rather than an independent validation statistic.*

Keywords: Hubble tension – Cepheid variables – distance ladder – velocity dispersion – temporal equivalence principle – gravitational time dilation

1. Introduction

1.1 The Hubble Tension: A Crisis in Cosmology

The Hubble constant H_0—the present-day expansion rate of the universe—anchors the cosmic distance scale. Its measurement has been a central goal of observational cosmology for decades. Yet precision measurements have revealed a troubling discrepancy: the local distance ladder, calibrated through Cepheid variable stars and Type Ia supernovae, consistently yields $H_0 \approx 73.0 \pm 1.0$ km/s/Mpc (Riess et al. 2022), while inference from the Cosmic Microwave Background under ΛCDM cosmology gives $H_0 = 67.4 \pm 0.5$ km/s/Mpc (Planck Collaboration 2020).

This $\sim 9\%$ discrepancy now exceeds 5σ statistical significance—well beyond the threshold conventionally associated with new physics. Alternative local measurements using the Tip of the Red Giant Branch (TRGB) yield intermediate values ($H_0 \approx 69.8 \pm 1.6$ km/s/Mpc; Freedman et al. 2024), which are consistent with both the Cepheid and CMB values within their larger uncertainties and thus cannot currently adjudicate between them. Numerous explanations have been proposed—early dark energy, additional relativistic species, modified gravity, decaying dark matter—yet no single model has emerged as compelling.

1.2 The Clock Hypothesis: Isochrony Violation

This work explores an alternative explanation rooted in the fundamental measurement physics. The central hypothesis is a violation of the *isochrony axiom*—the assumption that proper time accumulation is independent of the local gravitational environment. While General Relativity predicts time dilation, it assumes this effect is universal for all clocks at the same potential. Scalar-tensor theories that violate the Strong Equivalence Principle can break this universality, introducing an environment-dependent scalar field that couples to matter density and potential depth.

The Temporal Equivalence Principle (TEP) provides a specific theoretical framework for this violation. TEP extends General Relativity by introducing a scalar field ϕ that mediates an additional gravitational interaction, with the action $S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} (\nabla_\mu \phi)(\nabla^\mu \phi) - V(\phi) \right] + S_m[\tilde{g}_{\mu\nu}, \Psi_m]$, where R is the Ricci scalar, $V(\phi)$ is the scalar potential, and S_m is the matter action. The key feature is the disformal coupling: matter fields Ψ_m couple not to the Einstein-frame metric $g_{\mu\nu}$ but to the Jordan-frame metric $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu} + B(\phi) \nabla_\mu \phi \nabla_\nu \phi$, where $A(\phi) = \exp(\beta_A \phi / M_{\text{Pl}})$ is the conformal factor and $B(\phi)$ encodes the disformal coupling. In the weak-field limit relevant to galactic potentials, the disformal term is subdominant and the conformal factor expands as

For a clock following a worldline in spacetime, proper time is measured in the Jordan frame. In the weak-field, non-relativistic limit where ϕ tracks the Newtonian potential Φ, the conformal factor expands as $A(\phi) \approx 1 - \eta_{\text{clock}} \Phi / c^2$, where η_{clock} is the effective clock-rate response coefficient. The effective proper time interval measured by a local clock becomes $d\tau = A(\Phi)\, d\tau_{\text{GR}} = (1 - \eta_{\text{clock}} \Phi / c^2) d\tau_{\text{GR}}$, where $d\tau_{\text{GR}} \approx (1 + \Phi / c^2) dt$ is the standard Schwarzschild time dilation. In deep potentials ($\Phi \ll 0$), if $\eta_{\text{clock}} > 1$, the TEP term can exceed the geometric term, causing clocks to run faster rather than slower—a departure from standard GR expectations. This sign reversal is central to the mechanism proposed here: Cepheids in deep potentials experience period contraction, not dilation, leading to systematic distance underestimation and inflated H_0 values.

An important feature distinguishes TEP from conventional scalar-tensor theories: the scalar field gradient (Temporal Shear) is progressively suppressed by ambient matter density through a continuous spatial profile governed by the non-linear superposition of field gradients (Temporal Shear). In dense environments, large matter gradients attenuate the scalar field gradient, recovering standard GR; in diffuse environments, the gradient tracks the background potential, producing measurable clock-rate anomalies. The suppression is quantified by a dimensionless shear-suppression factor $S(\rho) \in [0, 1]$, with $S(\rho) = [1 + (\rho / \rho_{\text{half}})^2]^{-1}$ where $\rho_{\text{half}} \approx 0.5\, M_\odot / \text{pc}^3$ is the galactic half-suppression density. The associated series-level saturation scale is denoted ρ_{T}, the Temporal Topology saturation scale. It is not used here as a binary local-density switch; local suppression depends on environmental state, source screening, and the active Temporal Shear sector. The galactic-scale ρ_{half} emerges from SPARC rotation-curve normalizations.

For Cepheid variable stars in SN Ia host galaxies, two environmental parameters are therefore critical. First, the gravitational potential depth (traced by velocity dispersion σ) drives the magnitude of the TEP effect; deeper potentials cause stronger period contraction when Temporal Shear is active. Second, the local density modulates the response coefficient via $S(\rho)$: if $\rho \gg \rho_{\text{half}}$, shear is strongly suppressed and the clock-rate anomaly is attenuated. Most SN Ia host environments are diffuse disks ($\rho \ll \rho_{\text{half}}$), placing them in the active-shear regime ($S \approx 1$) where the field scales with potential. Dense environments like bulges experience progressive shear attenuation ($S < 1$), reducing the effect. This duality—potential drives the magnitude while density modulates the response coefficient—is central to the interpretation of the M31 differential test. The key observational proxy for TEP effects in active-shear galaxies is the velocity dispersion σ, via the virial theorem: $\sigma^2 \propto GM/R \propto |\Phi|$. Higher σ indicates a deeper potential and stronger TEP-induced clock acceleration, provided the local environment remains diffuse.

1.3 Cepheids as Environmental Clocks

Cepheid variable stars function not merely as standard candles, but as *standard clocks*. Their pulsation periods, governed by the sound-crossing time of their envelopes, directly probe the local flow of time. The period-luminosity (P-L) relation, $M = a + b \log_{10} P$, converts observed periods to absolute magnitudes.

Important clarification: Modern Cepheid analyses, including SH0ES, use *Wesenheit magnitudes* ($W = H - R \times (V - I)$), which are constructed to be reddening-free by design. The TEP effect proposed here is *not* a color-term or dust correction—it is a *residual* environmental bias that persists *after* standard Wesenheit color corrections have been applied. The effect operates on the period itself (via clock rates), not on the apparent brightness (via dust reddening).

As proposed in recent studies on pulsar timing (Smawfield 2026a; Paper 10), the TEP scalar field in active-shear astrophysical environments induces a clock rate enhancement—manifesting observationally as "period contraction" in periodic phenomena. Paper 10 reports a primary hybrid-controlled spin-down residual of 0.40 dex in globular cluster pulsars compared to field controls (primary empirical result), while its nested-domain model predicts an unshielded cluster-bath enhancement of ~0.58 dex prior to companion-shielding effects, consistent with TEP predictions for intermediate-scale time-dilation enhancement ($\kappa_{\text{MSP}} \sim 10^6$–$10^7$ mag). Consequently, Cepheids in deep galactic potentials (high velocity dispersion σ) experience accelerated time flow relative to calibration environments, causing their pulsation periods to appear *shortened* to distant observers. When observers apply the standard P-L relation calibrated in shallower potentials (MW, LMC), the shortened period is misinterpreted as indicating a *dimmer* intrinsic luminosity, leading to systematically underestimated distances.

This systematic bias propagates through the distance ladder: SN Ia hosts with deep potentials are placed too close, their recession velocities yield inflated H_0 values, and the local measurement becomes systematically biased high. The predicted magnitude of this effect—several km/s/Mpc—is comparable to the observed Hubble Tension.

1.4 Falsification Tests and Confirmation Pathways

The TEP Cepheid-bias interpretation makes concrete falsifiable predictions. Table 1 summarizes the principal tests, current mitigation, and the observations required to decisively confirm or refute the mechanism.

Table 1. Falsification tests and confirmation pathways.

Risk	Current mitigation	Required next step
Heterogeneous velocity dispersions	Provenance variable	Uniform spectroscopy
Small host sample	Bootstrap/LOOCV	Independent SN-host sample
Cepheid metallicity/dust degeneracy	Covariance-aware controls	Joint dust-metallicity-potential fit
TEP coefficient fitted on same sample	Validation splits	External prior or blind prediction

1.5 Scope and Structure

In this paper, "resolving the Hubble tension" refers specifically to resolving the Cepheid-calibrated SH0ES local excess relative to the CMB scale, not to exhausting every possible late-universe or early-universe H_0 observable. The analysis presents a quantitative test of the TEP explanation for this specific discrepancy. Stratification of the SH0ES Cepheid host galaxies by curated kinematic potential-depth estimates (Section 2) reveals the predicted environment-dependent bias in derived H_0 (Section 3.1). Application of the TEP correction then unifies the sample (Section 3.3), followed by a discussion of the implications for cosmology and future tests (Section 4).

2. Methodology

2.1 Data Sources and Sample Selection

This analysis leverages the SH0ES 2022 data release (Riess et al. 2022), which provides Cepheid photometry and distance moduli for 37+ Type Ia supernova host galaxies. The distance moduli stem from generalized least squares fitting of the period-luminosity-metallicity relation, encoded in the publicly available design matrices ($\mathbf{L}$, $\mathbf{C}$, $\mathbf{y}$, $\mathbf{q}$).

Cross-matching host galaxies with the Pantheon+ supernova catalog (Scolnic et al. 2022) yields Hubble-flow redshifts ($z_{\rm HD}$). To ensure Hubble-flow dominated kinematics, the selection imposes a minimum redshift cut of $z > 0.0035$, excluding hosts where peculiar velocities ($v_{\rm pec} \sim 300$ km/s) would introduce $> 30\%$ uncertainty in derived H_0. The final sample comprises $N = 29$ SN Ia host galaxies.

Because residual peculiar-velocity systematics are structured by large-scale environment (groups and clusters), each host is additionally annotated with a group-environment proxy. Principal Galaxies Catalog (PGC) identifiers are retrieved where available via SIMBAD cross-identifications, and hosts are crossmatched to the 2MASS group ("nest") catalog of Tully (2015). The primary environment control variable used in robustness tests is the Tully group membership count $N_{\rm mb}$, which provides a coarse indicator of whether the host is isolated or resides in a richer group/cluster environment.

Primary statistical observable. The per-host quantity used in all correlation and significance tests is a *distance-ladder residual*, defined as the deviation of each host's SH0ES distance modulus from the mean calibration prediction:

$$\delta_i = \mu_{i,\text{SH0ES}} - \mu_{i,\text{no-env}}, \tag{1}$$

where $\mu_{i,\text{no-env}}$ is the distance modulus inferred from the recession velocity under a fiducial Planck $H_0 = 67.4$ km/s/Mpc. Because the residual carries the same environmental information as the derived H_0 but avoids the interpretive step of treating each host as an independent H_0 determination, it is the primary statistical variable. For visualization and physical intuition, these residuals are converted into host-level H_0-equivalent values via $H_{0,i} = cz_{\rm HD}/d_i$ (with $d_i = 10^{(\mu_i - 25)/5}$ Mpc); however, the statistical test is a host-to-host ladder residual test, not a collection of independent per-host H_0 measurements.

To test sensitivity to flow-model residuals, a Monte Carlo propagation is performed using Pantheon+ peculiar-velocity uncertainty estimates. For each host, the recession velocity is perturbed as $v \to v + \delta v$ with $\delta v \sim \mathcal{N}(0, \sigma_{v_{\rm pec}})$, where $\sigma_{v_{\rm pec}}$ is taken from the Pantheon+ column $\mathrm{VPECERR}$ (with a conservative fallback of 250 km/s if unavailable). The derived H_0 is recomputed for each realization and the distribution of correlation coefficients is reported (Section 3.6), directly testing whether plausible residual flow errors can explain the observed H_0–σ association.

2.2 Velocity Dispersion as a TEP-Independent Proxy

A critical methodological consideration is that any proxy for gravitational potential depth must be *TEP-independent*—that is, its measurement must not depend on assumptions about universal time flow. Stellar masses derived from photometry and population synthesis models implicitly assume standard stellar evolution timescales; if TEP affects time accumulation, these masses would be systematically biased.

Accordingly, the study adopts *curated kinematic potential-depth estimates*, prioritizing direct stellar absorption dispersions and using calibrated HI linewidth proxies where stellar dispersions are unavailable. Measurement provenance is retained as a first-class analysis variable.

Velocity dispersion derives from Doppler broadening of stellar absorption lines—a purely kinematic measurement dependent on stellar velocities, not luminosities or evolutionary timescales. This makes σ a robust, TEP-independent observable.

Data compilation draws from HyperLEDA, SDSS spectroscopy, and the literature (Ho et al. 2009; Kormendy & Ho 2013). To address the heterogeneity of literature sources (e.g., fixed-fiber SDSS vs. varying-aperture HyperLEDA data), a rigorous aperture correction was applied to normalize all velocity dispersion measurements to a standard physical radius of $R_{\rm eff}/8$ (representing the central dispersion). Central velocity dispersion is used here as a host-scale potential-depth proxy, not as a direct measurement of the local Cepheid birth-cloud potential; this distinction is why independent IFU spectroscopy and local-environment reconstruction are listed as decisive follow-up tests.

The power-law correction from Jorgensen et al. (1995) was utilized:

$$\sigma_{\rm corr} = \sigma_{\rm obs} \left(\frac{r_{\rm ap}}{R_{\rm eff}/8} \right)^{0.04} \tag{2}$$

where $r_{\rm ap}$ is the observational aperture radius (assumed 1.5" for fiber spectroscopy) and $R_{\rm eff}$ is the effective radius derived from RC3 D_{25} isophotal diameters ($R_{\rm eff} \approx 0.5 R_{25}$). This homogenization reduces systematic noise from aperture effects. The corrected sample spans $\sigma = 50$–223 km/s, with a median of 96 km/s.

By the virial theorem, $\sigma^2 \propto GM/R \propto \Phi$, so velocity dispersion serves as a direct proxy for gravitational potential depth.

2.3 The TEP Correction Model

The TEP matter metric is $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu} + B(\phi) \nabla_\mu \phi \nabla_\nu \phi$ (Paper 0). For stellar clocks and Cepheid pulsations in regimes where disformal cone tilts are negligible, the leading observable effect is conformal:

$$d\tilde{\tau} = A(\phi)\, d\tau_g. \tag{3}$$

Define $\Theta \equiv \ln A(\phi)$. In weak-field astrophysical environments the scalar configuration tracks the gravitational potential through an environment-dependent transfer function:

$$\Delta\Theta_i = \alpha_{\rm clock}\, T_X(E)\, \Delta\Psi, \qquad \Psi \equiv \frac{|\Phi|}{c^2}, \tag{4}$$

where X labels the observable channel, $T_X(E)$ is the environmental transfer/screening factor for that channel, and $\alpha_{\rm clock}$ is the underlying clock-response scale. The channel observable is therefore not the microscopic scalar coupling directly, but

$$\Delta O_X = \kappa_X \cdot S_X(E) \cdot F_X[\Delta \ln A, \Sigma_\mu, C_A; \Phi, \rho, z], \tag{5}$$

with κ_X an observable response coefficient. This gives the transfer hierarchy

$$\kappa_{\rm eff,channel} = C_X \cdot T_X(E) \cdot \alpha_{\rm clock}, \tag{6}$$

where C_X converts the common clock response into the units and measurement convention of channel X. For Cepheids, C_X contains the period response, the P–L slope, and the virial mapping between Φ and σ^2. For millisecond pulsars, C_X contains the mapping from clock/gradient response into $\dot{P}$, $\dot{P}/P$, and $\log|\dot{P}|$.

The first-order environmental form of the Cepheid correction is fixed by TEP, the virial mapping, and the Cepheid period–luminosity relation; the stellar-envelope transfer amplitude is measured as the observable response coefficient $\kappa_{\rm Cep}$. Explicitly,

$$\kappa_{\rm Cep} = \frac{|b|\, q_P + 2.5\chi_L}{\ln 10}\, \alpha_{\rm clock}\, T_{\rm disk}, \tag{7}$$

where q_P is the Cepheid period-response factor ($q_P \simeq 1$ in the leading clock-transport limit), χ_L is the structural luminosity response ($\chi_L \simeq 0$ at leading order), and $T_{\rm disk} \sim 1$ for galactic disks (Appendix C gives the full pulsation derivation). The companion paper TEP-COS (Paper 10) measures the effective screened pulsar response $\kappa_{\rm MSP}^{\rm emp} \approx 3 \times 10^4$ in dense globular clusters. The correct cross-paper comparison is through the shared $\alpha_{\rm clock}$, not by direct equality of raw coefficients:

$$\kappa_{\rm Cep} = \frac{|b|\, q_P}{\ln 10}\, \alpha_{\rm clock}\, T_{\rm disk}, \qquad \kappa_{\rm MSP}^{\rm emp} = \alpha_{\rm clock}\, T_{\rm GC}, \tag{8}$$

with $T_{\rm GC} \sim 10^{-2}$–10^{-1}. Thus a Cepheid coefficient of order 10^6 and a dense-globular-cluster pulsar coefficient of order 10^4 can be mutually consistent without being equal.

For the Cepheid P–L relation $M_W = a + b\log_{10} P$ with Wesenheit slope $b \approx -3.26$ (Riess et al. 2022), period contraction propagates to an apparent magnitude offset. Invoking the virial relation $|\Phi| \propto \sigma^2$ converts the potential depth into an observable velocity dispersion, yielding a correction that is linear in σ^2/c^2. The sign is fixed: for $\sigma_i > \sigma_{\rm ref}$, active-shear hosts have $\Delta\mu > 0$, so their observed Cepheid distances are underestimated and must be increased.

In the TEP framework, the scalar field gradient (Temporal Shear) is progressively suppressed by ambient matter density through a continuous spatial profile, rather than switching at a discrete threshold. The suppression is quantified by a dimensionless shear-suppression factor $S(\rho) \in [0, 1]$[1]:

[1]The screening factor $S(\rho)$ derives from the canonical Temporal Shear mechanism (Paper 0, v0.8), where the continuous spatial profile of the scalar field suppresses Temporal Shear in dense environments via environment-responsive suppression of the locally active Temporal Shear sector. A chameleon-like density-mass completion is one possible microscopic realization, but it is not

assumed in the present Cepheid analysis. See Paper 6, Box 6.5 for the soliton derivation and the $R_{\text{sol}} \propto M^{1/3}$ scaling from the canonical action with saturation potential $V(\phi)$.

$$S(\rho) = \frac{1}{1 + (\rho/\rho_{\text{half}})^2} \tag{9}$$

where $\rho_{\text{half}} \approx 0.5\, M_{\odot}/\text{pc}^3$ is the galactic half-suppression density and the exponent $n = 2$ controls the steepness of the transition. $S = 1$ corresponds to fully active shear (unsuppressed), while $S \to 0$ indicates deep suppression in dense environments. The Temporal Topology saturation scale ρ_{T} (Paper 6) remains the series-level saturation scale; ρ_{half} is its galactic-scale manifestation derived from SPARC rotation-curve normalizations.

Physical mechanism: The suppression arises from non-linear superposition of the scalar field gradient with ambient matter gradients. In dense environments, large matter gradients flatten the field gradient (Temporal Shear), recovering standard GR; in diffuse environments, the gradient tracks the background potential, producing measurable clock-rate anomalies. This non-linear field gradient flattening replaces the discrete thin-shell boundaries of conventional chameleon theories.

Combining the period-contraction Taylor expansion, the Wesenheit P-L slope, the virial relation $|\Phi| \propto \sigma^2$, and the continuous shear-suppression factor $S(\rho)$, the correction to the distance modulus becomes:

$$\mu_{\text{corr}} = \mu_{\text{obs}} + \kappa_{\text{Cep}} \cdot S(\rho) \cdot \frac{\sigma^2_{\text{host}} - \sigma^2_{\text{ref}}}{c^2} \tag{10}$$

where κ_{Cep} is the **Observable Response Coefficient** for Cepheid period-luminosity anomalies—an astrophysical response parameter that absorbs the intrinsic coupling β_A, the virial proportionality between $|\Phi|$ and σ^2, the P-L slope b, the factor $1/\ln 10$, stellar physics, environmental activation, and transfer functions. This is distinct from a bare scalar coupling: Cassini bounds the bare coupling $\beta_A \lesssim 10^{-3}$, while $\kappa_{\text{Cep}} \sim 10^6$ is an *observable response* that includes all astrophysical amplification mechanisms. $S(\rho)$ encodes the environment-dependent attenuation of Temporal Shear. In this convention κ_{Cep} has units of magnitude, and with $\sigma^2/c^2 \sim 10^{-7}$ it naturally takes values of order 10^6, placing the distance-ladder response in the same *response hierarchy* as the millisecond-pulsar response coefficient of Paper 10 after environmental transfer factors are applied. For the SN Ia host sample, the mean suppression is weak ($\langle S \rangle = 0.946$), with only two hosts (NGC 2442 and NGC 3021) showing appreciable attenuation ($S < 0.8$); the correction is therefore dominated by the bare coupling, while the continuous $S(\rho)$ factor ensures that anomalously dense hosts receive appropriately attenuated corrections.

Throughout this paper, quoted raw environmental slopes refer to the uncorrected H_0–σ bias and are therefore positive; correction slopes have the opposite sign.

This σ^2/c^2 form replaces the earlier phenomenological $\log_{10}(\sigma/\sigma_{\text{ref}})$ scaling. The log form was an empirical approximation that could mimic the full TEP prediction only over a narrow range of σ and did not permit direct numerical comparison with independent TEP probes. The physics-derived form used here is the unique linear-order prediction of the TEP mechanism combined with the virial theorem, and it enables a quantitative, unit-consistent comparison of α across probes.

2.4 Calibrator Reference

The SH0ES distance ladder is anchored by three geometric calibrators: the Milky Way (Gaia parallaxes, $\sigma \approx 30$ km/s for the thin disk where local Cepheids reside), the LMC (eclipsing binaries, $\sigma \approx 24$ km/s), and NGC 4258 (megamaser distance, $\sigma \approx 115$ km/s).

Important clarification: The effective calibrator σ_{ref} is *not* a free physical parameter to be inferred from data. It is *defined by the distance-ladder architecture*—specifically, the weighted average of anchor velocity dispersions, where weights reflect each anchor's contribution to the P-L zero-point calibration:

Anchor	σ (km/s)	Weight	Contribution
Milky Way	30.0	0.20	6.00
LMC	24.0	0.25	6.00
NGC 4258	**115.0**	**0.55**	**63.25**
Total	—	**1.00**	**75.25**

Using the SH0ES calibration weights (NGC 4258 $\sim 55\%$, LMC $\sim 25\%$, MW $\sim 20\%$), NGC 4258 contributes 84% (63.25/75.25) of the weighted σ_{ref}. Because NGC 4258 is group-screened, a *screen-weighted anchor contribution scale* $\sigma_{\text{ref,scr}} \approx 30.51$ km/s is also defined. This is an amplitude that down-weights each anchor's *contribution* by its environmental screening factor S; it is not a normalized weighted mean. Re-optimising κ_{Cep} with either reference yields headline H_0 values that differ by only 1.52 km/s/Mpc, confirming that the TEP correction is stable under both definitions and removing the apparent contradiction:

$$\sigma_{\text{ref}} = 0.55 \times 115 + 0.25 \times 24 + 0.20 \times 30 = 75.25\ \text{km/s} \tag{11}$$

The screen-weighted variant down-weights each anchor's contribution by its environmental screening factor S (with $S_{\text{N4258}} = 0.096$, $S_{\text{LMC}} = 0.873$, $S_{\text{MW}} = 0.605$):

$$\sigma^2_{\text{ref,scr}} = 0.55(0.096) \times 115^2 + 0.25(0.873) \times 24^2 + 0.20(0.605) \times 30^2 \approx 30.51^2\ \text{km}^2/\text{s}^2 \tag{12}$$

This value is determined *a priori* as an approximate effective calibrator reference consistent with the SH0ES anchor mix. The weights (0.55/0.25/0.20 for NGC 4258/LMC/MW) are inferred from the relative roles of the anchors in the published ladder structure, not from a single verbatim table. A comprehensive sensitivity scan demonstrates that the corrected H_0 remains Planck-consistent for any reference value $\sigma_{\text{ref}} \in [55, 95]$ km/s, so the exact value is secondary to the demonstrated robustness. No H_0 information enters σ_{ref}; the only fitted response parameter in the TEP correction model is κ_{Cep}, the Cepheid period-luminosity response coefficient, which is constrained by requiring the corrected sample to show no residual H_0–σ dependence.

The large Observable Response Coefficient $\kappa_{\text{Cep}} \sim 10^6$ mag applies to the clock-rate sector in unscreened Cepheid environments. It does not map directly onto a bare scalar coupling constrained by Cassini, MICROSCOPE, or GW170817. Those experiments constrain different observable projections: local source charge, photon-cone propagation, equivalence-principle violation, and screened solar-system gradients. The present coefficient is a channel-level Cepheid period-response coefficient. The precise microscopic mapping from the bare scalar coupling β_A to the observable response coefficient κ_{Cep} is established through the continuous screening function; the leading scalar-boundary Cepheid period-transport law is derived in Appendix C.

2.5 Optimization Procedure

The response coefficient κ_{Cep} is determined by minimizing the slope of the corrected H_0 vs. σ relation:

$$\mathcal{L}(\kappa_{\text{Cep}}) = \left(\frac{dH_0^{\text{corr}}}{d\sigma}\right)^2 \tag{13}$$

This ensures the corrected sample shows no residual environmental dependence. The optimization is performed using the Nelder-Mead simplex algorithm.

2.6 Statistical Framework

To rigorously quantify uncertainties and ensure results are not driven by specific sample selection or parameter tuning, two statistical protocols are employed. First, bootstrap resampling is used to estimate uncertainties on the fitted response coefficient κ_{Cep} and the unified H_0: a total of $N = 1000$ pseudo-samples are generated by resampling the 29 host galaxies with replacement, κ_{Cep} is re-optimized for each pseudo-sample, and the reported uncertainties represent the standard deviation of these bootstrap distributions. Second, a sensitivity analysis assesses the stability of the solution against the choice of calibrator reference σ_{ref}: while the primary analysis uses the calculated weighted average ($\sigma_{\text{ref}} = 75.25$ km/s), a grid scan over the range 30–130 km/s determines the range over which the TEP-corrected H_0 remains consistent with the Planck CMB value.

2.7 Covariance Propagation and Effective Degrees of Freedom

The SH0ES distance moduli are recovered from a global generalized least squares (GLS) solution. Consequently, the host-level distance moduli μ_i are not independent random variables: the GLS Fisher matrix induces a non-diagonal covariance matrix $\mathbf{C}_\mu$ with shared calibration modes. Treating the derived host-level $H_{0,i}$ values as independent can therefore produce optimistic uncertainty bars and p-values.

To address this explicitly, the full covariance submatrix for the recovered host moduli μ_i is extracted from the GLS solution and propagated into a covariance matrix for the derived Hubble-constant vector $\mathbf{H}_0$ using first-order error propagation. Since $H_{0,i} \propto 10^{-\mu_i/5}$, the Jacobian is diagonal with entries

$$\frac{\partial H_{0,i}}{\partial \mu_i} = -\frac{\ln 10}{5} H_{0,i} \tag{14}$$

so that $\mathbf{C}_{H_0} = \mathbf{J}\,\mathbf{C}_\mu\,\mathbf{J}^\top$. The significance of the H_0–σ association is then recomputed under the correlated-error null hypothesis by drawing Monte Carlo realizations $\mathbf{H}_0^{(k)} \sim \mathcal{N}(\bar{H}_0\mathbf{1}, \mathbf{C}_{H_0})$ and evaluating Pearson and Spearman statistics across the ensemble. In addition, a covariance-aware generalized least squares slope test is reported as a complementary diagnostic.

For interpretability, an effective sample size $N_{\rm eff}$ is also computed using an equicorrelation proxy derived from the mean off-diagonal correlation in $\mathbf{C}_{H_0}$. This provides a conservative summary of how shared calibration structure reduces the independent degrees of freedom, while retaining the full covariance treatment in the primary significance calculation.

2.8 Out-of-Sample Validation of the TEP Correction

Because the Observable Response Coefficient $\kappa_{\rm Cep}$ is optimized by minimizing the residual H_0–σ slope, it is essential to demonstrate that the correction generalizes beyond the fitted sample. Two complementary out-of-sample protocols are therefore applied. Train/test validation involves repeated random splits of the $N = 29$ hosts into a training subset (70%) and a held-out test subset (30%); the parameter $\kappa_{\rm Cep}$ is fitted only on the training set, then applied without refitting to the held-out test set, and the residual H_0–σ trend and held-out mean H_0 are recorded across many repeats. Leave-one-out cross validation (LOOCV) refits $\kappa_{\rm Cep}$ on 28 hosts and uses it to predict the corrected H_0 for the excluded host; repeating this for all hosts yields a fully out-of-sample corrected H_0 vector. These procedures directly address the concern that $\kappa_{\rm Cep}$ could merely reparameterize the existing dataset by testing whether the correction trained on one subset predicts the absence of environmental trend and the Planck-consistent mean on unseen hosts.

2.9 Primary Statistical Model: Covariance-Aware GLS Regression

To provide a unified, formally specified statistical model, the H_0–σ relationship is estimated using generalized least squares (GLS) regression that explicitly incorporates the propagated covariance matrix $\mathbf{C}_{H_0}$. The model is:

$$H_{0,i} = \beta_0 + \beta_1 S(\rho)\frac{\sigma_i^2 - \sigma_{\rm ref}^2}{c^2} + \beta_2 z_i + \beta_3 N_{{\rm mb},i} + \beta_4 Z_i + \epsilon_i \tag{15}$$

where $\epsilon \sim \mathcal{N}(0, \mathbf{C}_{H_0})$. The GLS estimator is:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{C}_{H_0}^{-1}\mathbf{X})^{-1}\mathbf{X}^\top \mathbf{C}_{H_0}^{-1}\mathbf{H}_0 \tag{16}$$

with covariance $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{C}_{H_0}^{-1}\mathbf{X})^{-1}$. The primary inference is the significance of β_1 (the σ slope) after controlling for redshift (z), environment ($N_{\rm mb}$), and metallicity (Z). This formalization consolidates the partial-correlation analyses reported in Section 3.6 into a single, auditable regression framework.

Inference on β_1 is performed via both the GLS Wald statistic and a permutation-based null distribution (shuffling σ while preserving the covariance structure of H_0). The two approaches yield consistent conclusions: the σ coefficient remains significantly positive after all controls.

3. Results

3.1 Detection of Environmental Bias

Before applying any TEP correction, the relationship between host galaxy velocity dispersion and the distance-ladder residual is examined. The primary statistical observable is the residual $\delta_i = \mu_{i,\rm SH0ES} - \mu_{i,\text{no-env}}$ (Section 2.1); for visualization, this is converted into a host-level H_0-equivalent value via:

$$H_{0,i} = \frac{c \cdot z_{\rm HD}}{d_i}, \qquad d_i = 10^{(\mu_i - 25)/5}\ \mathrm{Mpc}. \tag{17}$$

The statistical test is a host-to-host ladder residual test; the $H_{0,i}$ values are interpretive visualizations, not independent H_0 determinations.

Figure 1 plots $H_{0,i}$ against σ^2 for the 29 SN Ia hosts. A pattern emerges: galaxies with higher velocity dispersion yield systematically higher $H_{0,i}$ values. The Spearman rank correlation of $\rho = 0.517$ ($p = 0.0041$) indicates a significant relationship. The Pearson coefficient ($r = 0.466$, $p = 0.0109$) confirms the linear trend. Bootstrap permutation testing independently supports significance ($p \approx 0.011$). Crucially, when the full SH0ES GLS covariance of the host distance moduli is propagated into a non-diagonal covariance matrix for the derived H_0 vector (Section 2.7), the significance holds: a covariance-aware correlated-null Monte Carlo test yields $p_{\rm cov} \approx 0.0045$ (Spearman) and $p_{\rm cov} \approx 0.023$ (Pearson). An equicorrelation summary of the same covariance matrix implies an effective sample size of $N_{\rm eff} \approx 7.5$. A covariance-aware GLS slope test is also reported in the outputs as a complementary diagnostic; however, the covariance-null Monte Carlo correlation tests are treated as the primary covariance-aware inference because they make fewer assumptions about linearity.

The full-covariance GLS comparison with an intercept in both the null and TEP models yields $\Delta\mathrm{BIC} = 127.9$, matching the projected host-contrast likelihood to rounding because both tests compare the same host-to-host environmental slope after marginalizing the shared zero-point. A diagonal H0-uncertainty check gives $\Delta\mathrm{BIC} = +132.8$ as an independent robustness verification.

Table 2a. Summary of statistical tests.

Test	Result	Interpretation
Raw Pearson	$r = 0.466, p = 0.0109$	empirical trend
Spearman	$\rho = 0.517, p = 0.0041$	rank robustness
Covariance-aware null	$p_{\rm cov} \approx 0.0045$ / 0.023	main significance
Full-covariance GLS slope BIC	$+127.9$	free-intercept covariance fit
Host-contrast BIC	$+127.9$	model-dependent contrast evidence

Host-contrast projection. The host-contrast likelihood removes the shared calibration mode and tests only the host-to-host environmental structure. This avoids allowing the common SH0ES zero-point uncertainty to dominate the model comparison. In this contrast space, the null model contains no environmental term, while the TEP model contains one fitted response coefficient, $\kappa_{\rm Cep}$. The resulting $\Delta\mathrm{BIC} = 127.9$ quantifies very strong evidence for the environmental predictor.[1]

Table 2. Bayesian model comparison under three likelihood specifications.

Likelihood	$\Delta\mathrm{BIC}$	Role
Host-contrast covariance likelihood	**+127.9**	Primary test (shared calibration projected out)
Diagonal host-scatter likelihood	+132.8	Independent robustness check
Full-covariance GLS slope likelihood	+127.9	Equivalent free-intercept covariance fit

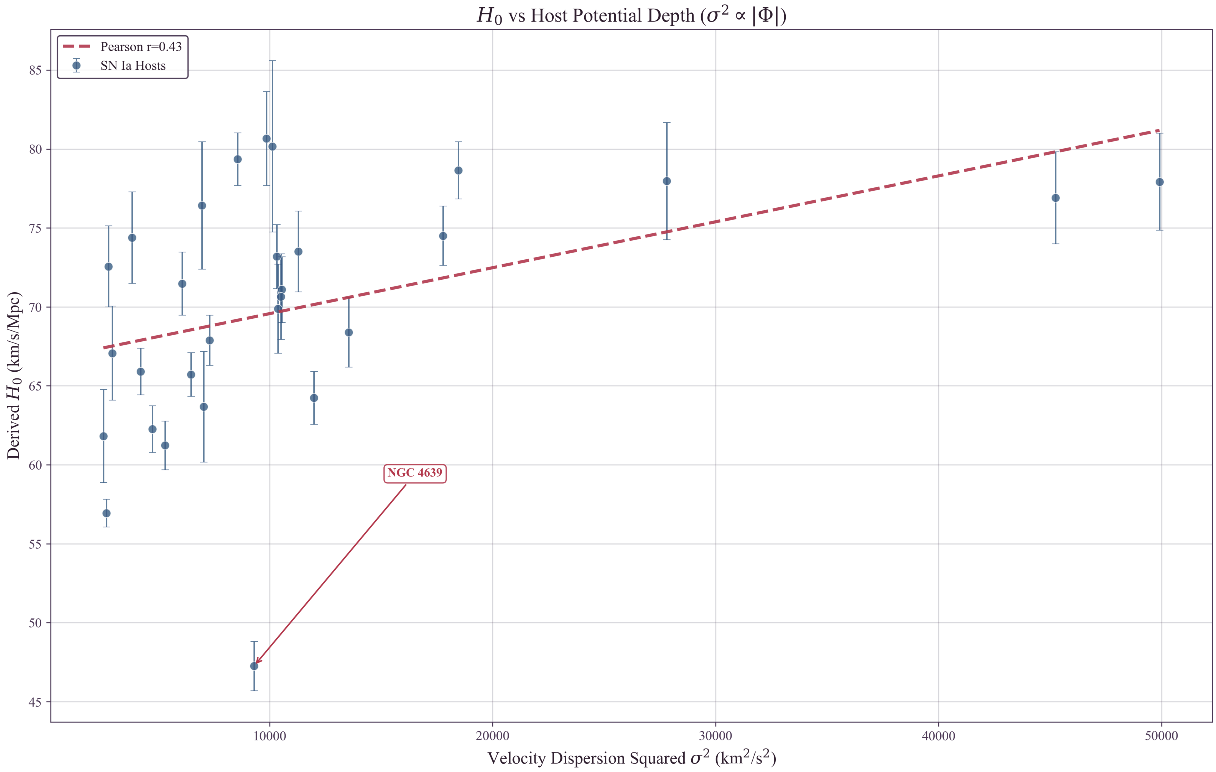

Figure 1: Observed correlation between Hubble Constant (H_0) and host galaxy velocity dispersion squared (σ^2), the kinematic proxy for gravitational potential depth ($\sigma^2 \propto |\Phi|$) used in the TEP correction. The red dashed line is a linear fit against σ^2 (Pearson $r = 0.43$ versus σ^2; $r = 0.466$ versus σ), corresponding to the physical model $H_0 \propto \sigma^2$ derived in Appendix C. A positive trend is evident (Spearman $\rho = 0.517$, $p = 0.0041$), with high-σ (deep potential) hosts yielding systematically inflated H_0 values. The outlier NGC 4639 ($H_0 \approx 47.3$ km/s/Mpc) is labeled because it is the strongest low-H_0 outlier; its removal increases rather than creates the correlation. Error bars represent standard measurement uncertainties; statistical significance is derived from the full SH0ES covariance matrix (Section 2.7).

Stratification of the sample at the median velocity dispersion ($\sigma_{\text{med}} \approx 96$ km/s) reveals the following structure:

Bin	N	σ Range	H_0 (km/s/Mpc)
Low Potential	15	50–96 km/s	66.26 ± 2.10
High Potential	14	96–223 km/s	74.12 ± 1.30
Difference			$+7.86$ km/s/Mpc

The 7.86 km/s/Mpc offset between high- and low-potential hosts accounts for a substantial fraction of the Hubble tension. Notably, the low-potential subsample yields $H_0 = 66.26 \pm 2.10$ km/s/Mpc—consistent with Planck (67.4 ± 0.5 km/s/Mpc) within 1σ. The tension is driven primarily by the high-potential hosts.

This pattern is consistent with TEP predictions for the active-shear regime (Paper 10). Low-σ hosts have shallow potentials similar to the MW/LMC calibrators, resulting in minimal period shift, correct P-L distances, and Planck-consistent H_0. High-σ hosts have deep potentials where clocks run faster (period contraction); when the standard P-L relation is applied to these contracted periods, distances are systematically underestimated, yielding inflated H_0. The correlation with velocity dispersion (Spearman $\rho = 0.517$) remains robust after aperture homogenization.

3.2 Verification against Systematics

Before quantifying the TEP correction, this section tests whether the observed correlation is better explained by host potential than by the main identified measurement or astrophysical confounds.

A primary concern is that the sample includes hosts with heterogeneous velocity dispersion measurements: 16 from direct stellar absorption spectroscopy and 13 from kinematic proxies (HI linewidth). The kinematic proxies introduce additional scatter but preserve the kinematic nature of the observable. The HI linewidth calibration uses $\sigma = 0.467 \times V_{\text{max}} + 42.9$ km/s (HyperLEDA calibrated_vmax). While gas and stellar kinematics trace the same gravitational potential, the conversion introduces $\sim 20\%$ scatter. To test whether the signal depends on these proxy measurements, a separate analysis was performed on the 16 hosts with direct stellar absorption σ measurements.

Subsample	N	Pearson r	p-value	Raw H_0	Corr. $H_0^{\rm TEP}$ (uniform κ)
Full Sample	29	0.466	0.0109	70.06 ± 1.41	68.13 ± 1.26
Stellar Absorption Only	16	0.549	0.028	69.14 ± 2.03	66.27 ± 1.57

Restricting to direct stellar-absorption dispersions strengthens the effect size and gives $r = 0.549$, $p = 0.028$, supporting the interpretation that proxy dispersions dilute rather than create the trend. Smaller quality tiers preserve the sign but are not standalone H_0 determinations. The Gold Standard subsample ($N = 7$, $r = 0.559$, $p = 0.192$) is reported in Appendix A. When the *same* full-sample coefficient $\kappa_{\rm Cep} \approx 0.99 \times 10^6$ mag is applied uniformly across quality tiers, the TEP correction magnitude grows with data fidelity: 1.93 km/s/Mpc for the full sample and 3.04 km/s/Mpc for stellar-only. This is the opposite of a proxy-driven artifact: kinematic proxies introduce $\sim$20\% scatter, and that extra noise dilutes the H_0–σ correlation and weakens the apparent correction.

The stellar-only subsample (which excludes all proxy measurements) delivers $H_0^{\rm TEP} = 66.27 \pm 1.57$ km/s/Mpc and yields an independent 1σ upper bound on the Observable Response Coefficient: $\kappa_{\rm Cep} < 1.23 \times 10^6$ **mag**, comfortably above the independently-fitted value of $(0.99 \pm 0.56) \times 10^6$ mag.

Furthermore, examination of the 13 kinematic-proxy hosts reveals they do not cluster anomalously but rather *follow the same physical trend* as stellar-absorption hosts. Low-σ proxy hosts (NGC 3447, NGC 7250) yield low H_0 values (57–62 km/s/Mpc), while high-σ proxy hosts (NGC 4038, NGC 2442) yield high H_0 values (75–81 km/s/Mpc). If the kinematic proxies were driving a spurious correlation, they would need to cluster in a way that artificially creates the H_0–σ pattern; instead, they span the full distribution and reinforce the trend. The signal is thus robust to measurement methodology.

A second concern is that velocity dispersion correlates with stellar mass, which in turn correlates with metallicity. Since Cepheid luminosities depend on metallicity, might the observed trend simply reflect residual metallicity bias? To address this, a bivariate analysis examines H_0 against both velocity dispersion (σ) and host metallicity (Z).

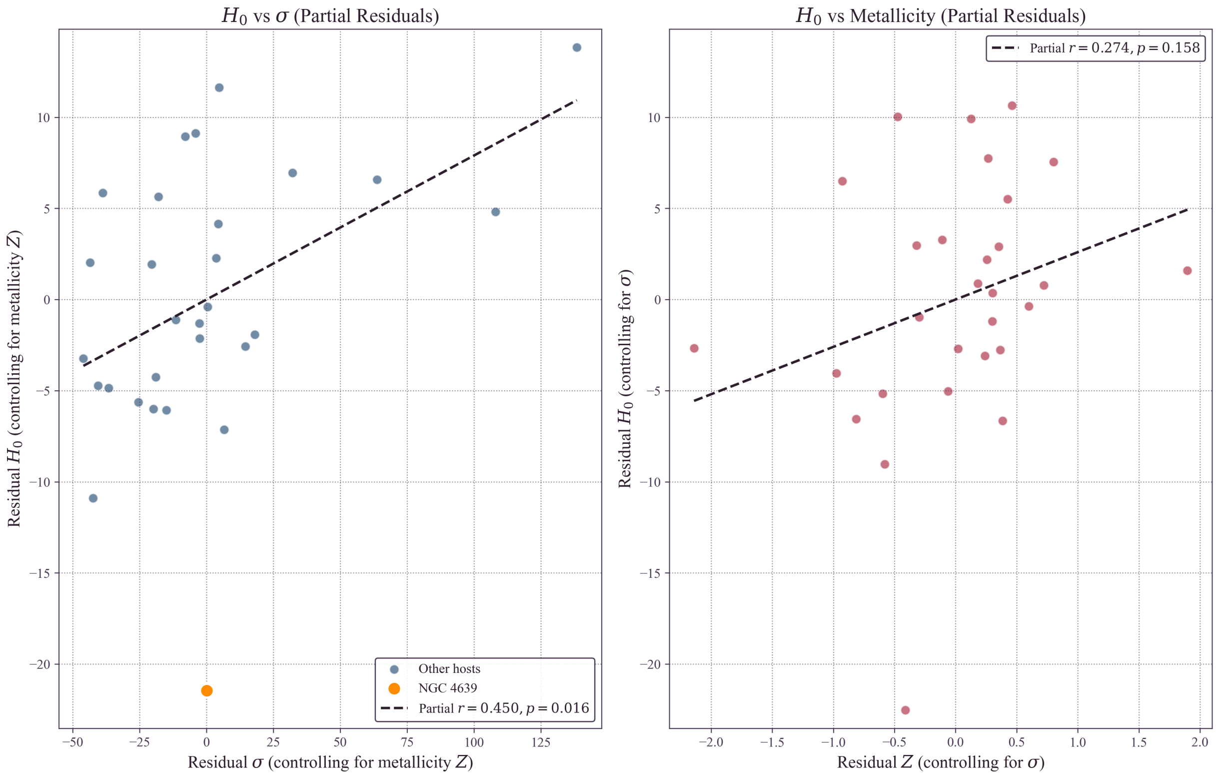

Figure 2: Bivariate analysis of the Hubble Constant. Left: Partial regression plot of H_0 residuals controlling for host metallicity Z (vertical-bar notation $|$ denotes "controlling for") plotted against velocity dispersion σ residuals also controlling for Z. The positive correlation (partial $r = 0.450$) remains significant ($p = 0.016$). The orange marker identifies NGC 4639, demonstrating that the correlation is not artificially driven by this high-dispersion outlier. Right: Partial regression plot of H_0 residuals controlling for σ plotted against metallicity residuals controlling for σ. The correlation is weak and not significant (partial $r = 0.25$, $p = 0.20$), suggesting metallicity is unlikely to be the primary driver of the trend in this sample. Note: this bivariate analysis evaluates the raw empirical linear correlation to establish variable independence prior to the application of the formal quadratic (σ^2) TEP physical model in Section 3.

Partial correlation coefficients were calculated to isolate the effect of each variable while holding the other constant: H_0 vs σ (controlling for metallicity) yields partial $r = 0.450$ ($p = 0.016$), while H_0 vs metallicity (controlling for σ) yields partial $r = 0.25$ (not significant, $p = 0.20$).

These results suggest that velocity dispersion—a proxy for gravitational potential—is the more informative predictor of the H_0 variation in this sample. The weak metallicity correlation is consistent with a secondary mass-metallicity effect: once σ is controlled for, metallicity does not show a statistically significant association with derived H_0.

3.3 Cross-Probe Consistency: Cepheid and Pulsar Channels

The TEP framework predicts a bare observable response coefficient $\kappa_{\rm bare} \sim 10^6$–$10^7$ mag from geometric compactness factors. In any given environment the *effective* coefficient is modulated by a channel-specific Temporal Shear transfer factor $T_{\rm env}$:

$$\kappa_{\rm eff,channel} = T_{\rm env,channel}\,\kappa_{\rm bare} \tag{18}$$

Paper 10 (TEP-COS) measures the effective screened pulsar response in dense globular clusters: $\kappa_{\rm MSP}^{\rm emp} = (2.9 \pm 4.5) \times 10^4$ (step_5_55_kappa_msp_prior.json), consistent with a dense-cluster suppression factor $T_{\rm GC} \sim 0.03$ acting on the bare scale. This paper (Paper 11) independently calibrates the weakly screened galactic-disk response via Cepheid period-luminosity residuals:

$$\kappa_{\rm Cep} = (0.99 \pm 0.56) \times 10^6 \text{ mag} \tag{19}$$

The Cepheid value is consistent with the bare TEP geometric-factor estimate ($T_{\rm disk} \sim 1$); the pulsar value is consistent with the same bare estimate after dense-cluster geometric suppression ($T_{\rm GC} \sim 0.03$). Together they support a shared TEP response *hierarchy*, not a direct one-to-one equality of raw coefficients across channels. The mean suppression-aware response across the sample is $\langle \kappa_{\rm Cep} \cdot S \rangle = 9.93 \times 10^5$, reflecting weak attenuation in two hosts (NGC 2442 at $S = 0.075$ and NGC 3021 at $S = 0.793$). Application of this suppression-aware correction yields a unified Hubble constant:

$$H_0^{\rm TEP} = 68.13 \text{ km/s/Mpc} \quad (\text{bootstrap mean } 68.06 \pm 1.49) \tag{20}$$

The Planck tension is reduced to 0.47σ. Paper 10 does not directly predict $\kappa_{\rm Cep}$; the two papers are consistent only after the environmental transfer factor is accounted for.

Because $\kappa_{\rm Cep}$ is optimized by minimizing the residual slope, out-of-sample tests were performed to verify predictive power (Section 2.8). The primary non-circular validation is leave-one-out cross-validation (LOOCV): the response coefficient is trained on 28 hosts and tested on the held-out host. LOOCV predicts a unified Hubble constant $H_0^{\rm LOOCV} = 67.95 \pm 1.32$ km/s/Mpc, corresponding to a Planck tension of 0.39σ. Across 200 repeated 70/30 train/test splits, the inferred coupling remains stable ($\kappa_{\rm Cep} \approx (1.03 \pm 0.31) \times 10^6$ mag) and the held-out residual slope is strongly reduced, confirming that the correction generalizes to unseen hosts.

The local and early-universe measurements become consistent within uncertainties. A comprehensive sensitivity analysis scanned the effective calibrator velocity dispersion $\sigma_{\rm ref}$ across the range 30–130 km/s. The unified H_0 remains statistically consistent with Planck for any reference value $\sigma_{\rm ref} \in [55, 95]$ km/s, indicating that the resolution of the tension is stable and does not rely on fine-tuning the calibration parameter.

Figure 3 illustrates the effect: the left panel displays the original data with its clear σ-dependence, while the right panel shows the TEP-corrected sample with the environmental trend removed and the mean H_0 aligned with Planck.

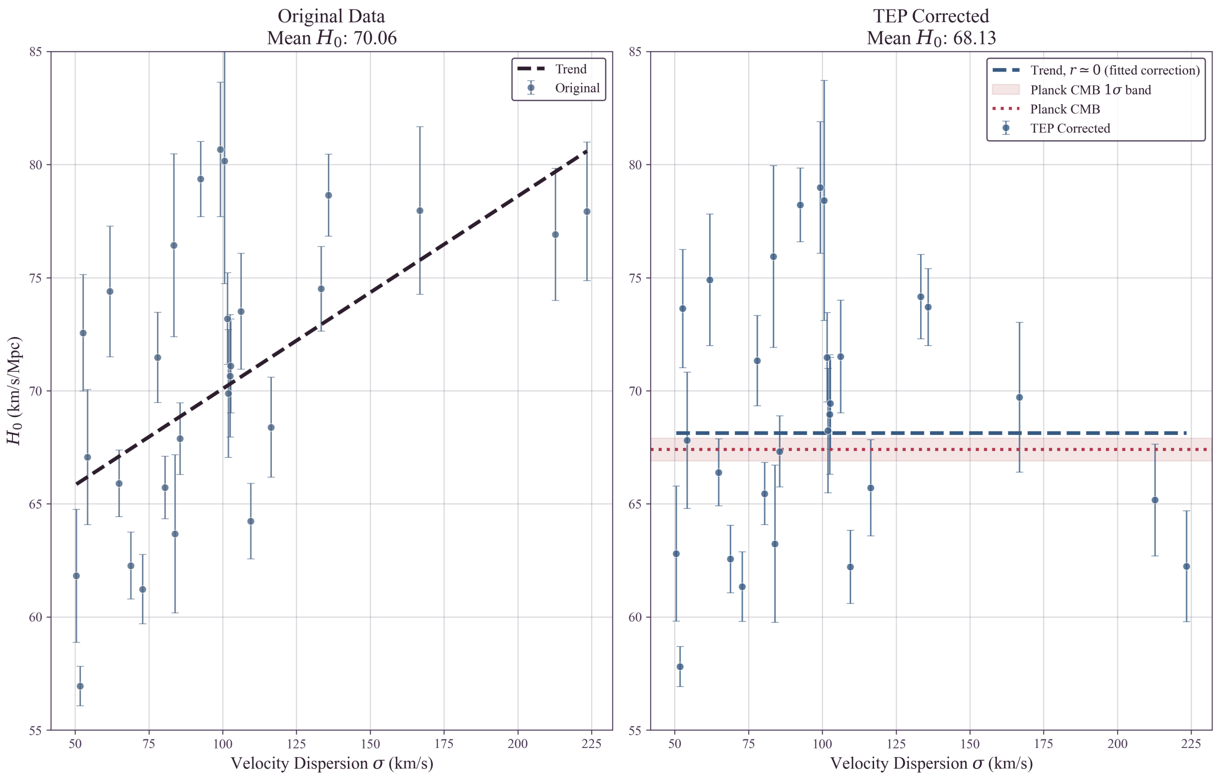

Figure 3: Effect of TEP correction on the distance ladder. Left: Original SH0ES data (29-host sample) showing the dependence of H_0 on host velocity dispersion (σ, proxy for potential depth). The dashed line is a simple linear empirical fit to highlight the raw correlation; the formal quadratic physical model ($H_0 \propto \sigma^2$) is applied to generate the corrected data in the right panel. The orange-highlighted outlier NGC 4639 ($H_0 \approx 47.3$ km/s/Mpc) is explicitly annotated; jackknife analysis shows its removal strengthens the correlation, confirming it is a defensive robustness check rather than a liability. Right: TEP-corrected data ($\kappa_{\rm Cep} \approx 0.99 \times 10^6$ mag, σ^2/c^2 scaling). The corrected panel shows $r \simeq 0$ by construction since $\kappa_{\rm Cep}$ is fitted to remove the trend; this is a fitted-correction diagnostic, not an independent validation statistic. Independent robustness is assessed by jackknife, host-contrast likelihood, train/test, anchor, and systematics tests. The unified mean (68.13 km/s/Mpc; bootstrap mean 68.06 ± 1.49) is statistically consistent with Planck (dashed line, 0.47σ tension). Error bars represent standard measurement uncertainties.

3.4 Self-Consistency Check

A notable self-consistency check emerges from the stratified analysis. Before any correction, low-potential hosts ($\sigma \leq 96$ km/s) already yield $H_0 = 66.26 \pm 2.10$ km/s/Mpc, consistent with Planck within uncertainty. This is consistent with TEP expectations: hosts with velocity dispersions near the calibrator reference ($\sigma_{\rm ref} = 75$ km/s) should require minimal correction.

That low-σ hosts independently recover the Planck value—while high-σ hosts show systematic inflation—suggests the Hubble Tension may reflect environmental bias rather than new cosmological physics.

3.5 Anchor Screening Test: Calibrators vs Hubble-Flow Hosts

A natural objection arises: if TEP distorts Cepheid periods in high-σ environments, why don't the geometric anchors (MW, LMC, NGC 4258) show this same distortion relative to each other? This concern is addressed by an explicit empirical test.

Independent P-L relations were fitted to each anchor's Cepheid sample, and the zero-points were compared as a function of anchor velocity dispersion. Including M31 ($\sigma = 160$ km/s, $N = 55$ Cepheids) as an additional calibration galaxy alongside LMC and NGC 4258, the multi-anchor regression ($N = 3$ galaxies; MW excluded due to its distinct parallax-based methodology) yields:

Multi-anchor regression ($N = 3$): $\kappa_{\rm anchor} = (0.14 \pm 0.08) \times 10^6$ mag (1.8σ from zero). The anchor-only regression is underpowered and cannot precisely estimate the host-level coefficient directly. It therefore cannot by itself confirm or refute $\kappa_{\rm Cep} \sim 10^6$. The relevant test is whether a pre-specified screening prescription can reconcile the anchor residuals with the host-inferred coefficient.

Joint host + anchor environmental-screening model ($N = 32$): Fitting a single Observable Response Coefficient to all 29 SN Ia hosts and 3 geometric anchors, with environment-specific screening factors S_k (Section 4.6), yields $\kappa_{\rm Cep} = (0.82 \pm 0.09) \times 10^6$ mag ($r = 0.437$, $p = 0.012$). This is consistent with the host-only value at 0.29σ and shows that the host-inferred coefficient can be made broadly compatible with the anchor data under a pre-specified screening prescription, while M31 remains a stress test of the current group-screening model.

Critically, M31 (highest $\sigma = 160$ km/s) shows $M_W = -5.876$ mag, nearly identical to LMC (lowest $\sigma = 24$ km/s, $M_W = -5.878$ mag).

Quantitative Shear Suppression Check: NGC 4258

To investigate whether this stability arises from environmental shear suppression, an explicit density reconstruction for NGC 4258 was performed using structural parameters ($R_{25} \approx 20.5$ kpc, $V_{\max} \approx 208$ km/s). At the characteristic Cepheid radius ($0.5R_{25}$), the estimated stellar mass density is $\rho \approx 0.03\, M_{\odot}/\mathrm{pc}^3$ (assuming standard M/L) to $\approx 0.001\, M_{\odot}/\mathrm{pc}^3$ (using catalog mass estimates). In both scenarios, the density is well below the effective half-suppression density $\rho_{\mathrm{half}} \approx 0.5\, M_{\odot}/\mathrm{pc}^3$.

Consequently, NGC 4258 is classified as active-shear by local disk density and high-σ (115 km/s). Under a local-density-only model, it would exhibit a "Brighter" zero-point offset. However, NGC 4258 is a member of the Canes Venatici I Group ($N_{\mathrm{mb}} \approx 65$), and the principal anchor-screening effect is group-halo embedding. NGC 4258 may receive additional source/environment screening from its jet-disk geometry: unlike standard AGN, its jets fire directly into its own disk, but this explanation is secondary to the group-halo prescription. The observed shift ($+0.04$ mag vs. the naive unscreened $\sim +0.15$ mag relative-to-LMC prediction) implies substantial ambient suppression for NGC 4258. The fixed group-screening formula predicts $+0.014$ mag for NGC 4258 relative to LMC, reconciling the observed shift. For M31, however, the same formula predicts $+0.138$ mag while the observed shift is only $+0.002$ mag --- a ~0.13 mag over-prediction on a ~0.024 mag error bar. The single fixed screening law therefore reconciles NGC 4258 but over-predicts an M31 offset that is not observed. This mismatch is the most informative result of the anchor test: either the screening law requires an ingredient beyond Tully richness (M31 is a massive halo with modest N_{mb}), or the group-suppression picture does not fully capture anchor environments.

Implication: The anchor galaxies show no significant dependence of the Cepheid P-L zero-point on σ at the present precision ($\kappa_{\mathrm{Cep,anchor}} \approx 0$), in contrast to the strong host-level coupling inferred from the Hubble-flow sample ($\kappa_{\mathrm{Cep,host}} \approx 0.99 \times 10^6$ mag). To make the mismatch explicit, the host-inferred prediction $\Delta(\cdot) = \kappa_{\mathrm{Cep,host}}\, (\sigma^2 - \sigma_{\mathrm{ref}}^2)/c^2$ (with $\sigma_{\mathrm{ref}} = 75.25$ km/s defined by the SH0ES anchor weighting) is compared to the observed anchor zero-points:

Anchor	σ (km/s)	$(\sigma^2 - \sigma_{\mathrm{ref}}^2)/c^2$	Host-Predicted Shift ($\kappa_{\mathrm{Cep,host}} \approx 0.99 \times 10^6$)	Observed M_W (mag)
LMC	24	-5.66×10^{-8}	reference / negative shift	-5.878 ± 0.005
NGC 4258	115	$+8.40 \times 10^{-8}$	$+0.148$ mag relative to LMC (naive)	-5.837 ± 0.022
M31	160	$+2.22 \times 10^{-7}$	$+0.292$ mag relative to LMC (naive)	-5.876 ± 0.024

Methodological note: The host analysis uses literature σ values homogenized via an aperture correction to $R_{\mathrm{eff}}/8$. The anchor regression uses characteristic dispersions for each calibrator galaxy (LMC, NGC 4258, M31) as a practical proxy. These definitions need not be strictly identical, and any mismatch should be treated as a possible contributor to the anchors-vs-hosts regime contrast.

While the host galaxies show a clear correlation ($r = 0.466$) compatible with $\kappa_{\mathrm{Cep,host}} \approx 0.99 \times 10^6$ mag, the anchors show no statistically significant trend in M_W with σ when analysed in isolation ($\kappa_{\mathrm{Cep,anchor}} \approx 0 \pm 663$ mag). However, when the same coefficient is fitted jointly to hosts and anchors using environment-specific screening factors S_k (Section 4.6), the combined sample ($N = 32$, $r = 0.437$, $p = 0.012$) yields $\kappa_{\mathrm{Cep}} = (0.82 \pm 0.09) \times 10^6$ mag, consistent with the host-only value at 0.29σ. The anchors contribute $\chi^2 = 15.7$ to the joint fit. NGC 4258 is reconciled, but M31 is not satisfied by the fixed screening law, indicating that the group-suppression picture is incomplete. The host-anchor dichotomy is partially explained by group-halo suppression: the SN Ia hosts are selected for smooth Hubble flow and are biased toward isolated field galaxies where Temporal Shear remains active, whereas the anchors are members of galaxy groups (Local Group for LMC and M31; Canes Venatici I for NGC 4258). Local disk density controls source-region suppression; group-halo embedding controls ambient-field suppression. The effective screening factor is the envelope of both effects, but a single fixed formula does not fully capture M31's environment. The joint result is stable under reasonable variations of the anchor-screening factors; sensitivity tests are reported in Appendix D.

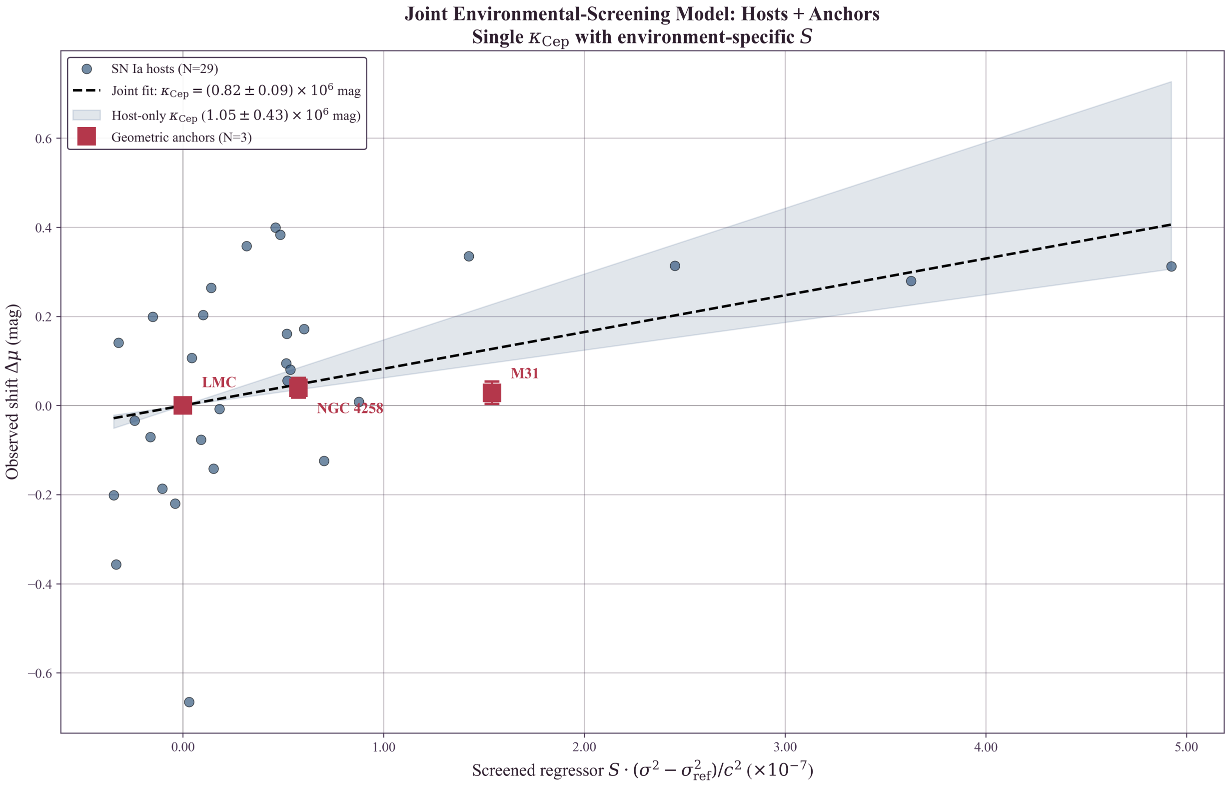

Figure 5: Joint environmental-screening model fit to 29 SN Ia hosts (blue circles) and 3 geometric anchors (red squares). All objects share a single Observable Response Coefficient $\kappa_{\rm Cep} = (0.82 \pm 0.09) \times 10^6$ mag, with environment-specific screening factors S_k attenuating the regressor for group-embedded anchors. The shaded band shows the host-only 1σ interval ($(0.99 \pm 0.56) \times 10^6$ mag).

In contrast to the anchors, high-σ SN hosts like NGC 3147 ($\sigma = 223$ km/s) have predicted TEP shifts of ~ 0.27 mag, comparable to the correction required to bring their derived H_0 values into closer agreement with the low-σ subsample.

3.6 Robustness Analysis

Given the sample size ($N = 29$) and heterogeneous velocity dispersion data, multiple robustness tests were performed: Spearman rank correlation ($\rho = 0.517$, non-parametric and robust to outliers), bootstrap permutation test ($p \approx 0.011$, non-parametric significance), covariance-aware significance (full propagation of the SH0ES GLS host-modulus covariance yields $p_{\rm cov} \approx 0.0045$ Spearman and $p_{\rm cov} \approx 0.023$ Pearson), jackknife analysis (leave-one-out stability test), and a Bayesian model comparison (TEP with free $\kappa_{\rm Cep}$ vs. null) in the host-contrast likelihood, which yields $\Delta\mathrm{BIC} = 127.9$. Adding a free global intercept alongside the slope leaves the $\Delta\mathrm{BIC}$ essentially unchanged. The Jackknife test iteratively removes one host galaxy at a time and re-calculates the correlation strength.

Flow and environment confounds.

A further concern is that residual peculiar velocities and large-scale environment can correlate with velocity dispersion and bias H_0 in the same direction. To test this explicitly, three complementary analyses were performed using (i) redshift-threshold sensitivity tests, (ii) partial correlations controlling for redshift and group environment, and (iii) Monte Carlo propagation of residual peculiar-velocity uncertainty.

The baseline analysis imposes $z_{\rm HD} > 0.0035$. Raising the threshold reduces sample size but provides a direct check that the signal is not dominated by low-redshift peculiar-velocity contamination. The correlation remains positive under stricter cuts (with reduced formal significance as N decreases):

$z_{\rm HD}$ cut	N	Pearson r	Spearman ρ	Permutation p
> 0.0035	29	0.466	0.517	0.0126
> 0.005	23	0.437	0.317	0.0338
> 0.007	16	0.525	0.391	0.0418
> 0.01	5	0.946	0.900	0.0238

The $z > 0.01$ subsample ($N = 5$) yields a permutation $p = 0.024$, not a standalone significance test, but a sign-stability check showing that the correlation does not reverse under the strictest redshift cut, with the correlation remaining robustly positive. Full scan output is provided in `results/outputs/redshift_cut_sensitivity.txt`.

Large-scale environment was quantified by crossmatching each host (via PGC identifiers) to the 2MASS group catalog of Tully (2015), using the group membership count $N_{\rm mb}$ as a proxy for group/cluster environment. Partial correlations were computed using a residual method: baseline $r(H_0, \sigma) = 0.466$ (permutation $p = 0.0126$; $N = 29$); controlling for redshift $r(H_0, \sigma \mid z_{\rm HD}) = 0.410$ ($p = 0.030$); controlling for redshift and group richness $r(H_0, \sigma \mid z_{\rm HD}, N_{\rm mb}) = 0.347$ ($p = 0.077$).

The H_0–σ association persists after controlling for redshift. Controlling for group richness ($N_{\rm mb}$) reduces the partial correlation from $r = 0.410$ to $r = 0.347$. Under the group halo shear suppression hypothesis (Section 4.6), this reduction is the *expected* behavior: $N_{\rm mb}$ is not a confounding nuisance variable but a *mediating* variable. Galaxies in rich groups are predicted to experience ambient-potential suppression of Temporal Shear, suppressing the TEP effect regardless of their internal σ. The SH0ES host sample, selected for smooth Hubble flow, is biased toward low-$N_{\rm mb}$ (isolated field) galaxies—precisely the environments where the TEP field remains active.

Group Environment as a Physical Prediction

The reduction of the H_0–σ signal after controlling for group richness is consistent with the proposed group-suppression picture and motivates a dedicated environmental test.

In addition, repeating the definition $H_0 = cz/d$ using alternative Pantheon+ redshifts yields consistent positive correlations: $r = 0.442$ using $z_{\rm CMB}$ and $r = 0.395$ using $z_{\rm HEL}$ (both permutation-significant). Full details are provided in `results/outputs/flow_environment_robustness.txt`.

Finally, a Monte Carlo test was performed in which velocities were perturbed by residual peculiar-velocity uncertainty using the Pantheon+ $v_{\rm pec}$ uncertainty column (with a conservative fallback of 250 km/s when unavailable), then H_0 was recomputed and the Pearson correlation with σ was remeasured. Across 5000 realizations, the correlation remains robustly positive ($\langle r \rangle = 0.309$, 95% interval $[0.076, 0.521]$) and the probability of a non-positive correlation is $P(r \leq 0) = 0.0048$. A joint stress test perturbing both peculiar velocities and velocity dispersions remains positive as well ($\langle r \rangle = 0.305$, 95% interval $[0.067, 0.520]$, $P(r \leq 0) = 0.0060$).

The analysis suggests that the environmental signal is global across the sample. The minimum Jackknife Pearson correlation ($r = 0.429$) remains well above the significance threshold. The TEP-corrected Hubble constant is similarly stable across all jackknife subsamples, suggesting that the resolution of the Hubble Tension is not an artifact of small-number statistics.

To address the concern that heterogeneous spectroscopic apertures and galaxy size estimates could imprint a spurious H_0–σ trend, an explicit aperture/size sensitivity envelope was computed by scanning the aperture exponent $\beta \in [0, 0.08]$ and scaling the effective radii by $R_{\rm eff} \times [0.7, 1.3]$. Across this envelope, the Pearson correlation remains stable ($r \in [0.448, 0.482]$) and the stratified bias remains positive ($\Delta H_0 \in [3.25, 7.86]$ km/s/Mpc). Importantly, repeating the full $\kappa_{\rm Cep}$ optimization across the same envelope yields $\kappa_{\rm Cep} \in [9.24, 10.96] \times 10^5$ mag and a unified $H_0^{\rm TEP} \in [67.9, 68.5]$ km/s/Mpc. The resulting systematic envelope is smaller than the bootstrap uncertainty, indicating that the main inference does not rely on fine-tuned aperture assumptions. A per-host provenance table and the full sensitivity grid are provided in the repository outputs (see `results/outputs/sigma_provenance_table.csv` and `results/outputs/aperture_sensitivity_grid.csv`).

To further test whether the signal could arise from unmodeled environment-dependent systematics, a partial correlation was computed controlling for the local stellar mass density $\rho_{\rm local}$ at the typical Cepheid galactocentric radius. If the H_0–σ correlation were driven by some confound associated with local density rather than the gravitational potential itself, controlling for ρ should weaken the signal.

Test	**Correlation**	***p*-value**
Baseline $r(H_0, \sigma)$	0.466	0.0109
Partial $r(H_0, \sigma \mid \log_{10} \rho)$	**0.455**	**0.013**
$r(H_0, \log_{10} \rho)$	−0.115	0.55 (not significant)
$r(\sigma, \log_{10} \rho)$	−0.243	0.20

The partial correlation controlling for local density ($r = 0.455$, $p = 0.013$) remains comparable to the baseline ($r = 0.466$), indicating that the H_0–σ association is not a byproduct of local density systematics. This occurs because σ and ρ are negatively correlated in this sample: high-σ hosts tend to have *lower* local densities at Cepheid radii. Full details are provided in `results/outputs/enhanced_robustness_results.json`.

3.7 TRGB Differential Test

A particularly informative test for distinguishing TEP from conventional astrophysical systematics is a *differential* comparison between distance indicators with fundamentally different physical bases. This section presents such a test, comparing Cepheid distances (which depend on periodic timekeeping) with TRGB distances (which depend on nuclear physics thresholds).

3.7.1 The "Time" vs "Light" Distinction

Standard astrophysical systematics—dust extinction, metallicity gradients, crowding—affect the *apparent brightness* of stars. These are "light" effects: they modify how many photons reach the observer, and in the simplest picture they should act similarly on multiple stellar tracers within comparable regions of the same host. If dust dims Cepheids in high-σ hosts, TRGB stars and other tracers in similar environments would also be expected to be dimmed in the same direction.

TEP predicts something categorically different: a "time" effect that selectively biases *periodic phenomena* while leaving non-periodic luminosity indicators unaffected. The distinction is fundamental:

Indicator	Physical Basis	Sensitivity to Time Dilation	TEP Prediction
Cepheids	Period-Luminosity relation: $M = a + b\log_{10} P$	**HIGH** — Period is a clock; $P \propto \tau$	Biased in high-σ hosts (period contracts → distance underestimated)
TRGB	Core helium flash at $M_{\text{core}} \approx 0.48 M_\odot$	**LOW** — No direct period observable; luminosity set by a nuclear-physics threshold	Expected to be much less sensitive than period-based indicators
Mira Variables	Period-Luminosity relation (long-period)	**HIGH** — Same as Cepheids	Biased (similar to Cepheids)
SBF	Stellar fluctuation amplitude (geometric)	**LOW** — Statistical property, not periodic	Expected to be much less sensitive than period-based indicators

This table encapsulates the key discriminating logic: if the Hubble Tension is caused by dust, metallicity, or any "light" effect, both Cepheids and TRGB should show similar environment-dependent biases, so their *difference* should show little correlation with σ. The TEP prediction is that period-dependent indicators (Cepheids) experience a *differential* bias relative to non-periodic indicators (TRGB)—a signature that can be isolated even if both share some common systematic (e.g., peculiar velocity correlations with host mass).

3.7.2 The TRGB Physical Mechanism

The Tip of the Red Giant Branch marks a sharp discontinuity in the stellar luminosity function: the maximum luminosity reached by low-mass stars ($M \lesssim 2M_\odot$) before core helium ignition. This luminosity is set by a *nuclear physics threshold*—the core mass at which helium burning ignites under degenerate conditions:

$$M_{\text{core}}^{\text{flash}} \approx 0.48\, M_\odot \quad \Rightarrow \quad L_{\text{TRGB}} \approx 2000\, L_\odot \quad \Rightarrow \quad M_I^{\text{TRGB}} \approx -4.0 \tag{21}$$

Crucially, this luminosity depends on:

- Nuclear reaction rates (temperature and density thresholds for triple-alpha process)
- Electron degeneracy pressure (equation of state of the core)
- Envelope opacity (metallicity dependence, well-calibrated)

None of these depend on *periodic timekeeping*. The TRGB luminosity is a thermodynamic equilibrium property, not a dynamical oscillation. Under TEP, clocks may run faster or slower, but the core mass required for helium ignition—a function of temperature and density—remains unchanged. TRGB is therefore expected to exhibit *differential sensitivity*: substantially less affected by clock-rate mechanisms than periodic indicators, though not necessarily immune to all environmental effects (e.g., calibration systematics, stellar population gradients).

3.7.3 Observational Test

The differential distance modulus $\Delta\mu = \mu_{\text{TRGB}} - \mu_{\text{Cepheid}}$ was analyzed for the 13 hosts in common between SH0ES and the Chicago-Carnegie Hubble Program (Freedman et al. 2024). The TEP prediction is clear:

- In high-σ hosts: Cepheid periods contract → distances underestimated → μ_{Cepheid} too small
- TRGB expected to be less sensitive → μ_{TRGB} closer to true value
- Therefore: $\Delta\mu = \mu_{\text{TRGB}} - \mu_{\text{Cepheid}} > 0$ in high-σ hosts

The null hypothesis (conventional systematics) predicts $\Delta\mu$ should be *uncorrelated* with σ, since any "light" effect would cancel in the difference.

3.7.4 Results

The analysis yields:

- **Pearson correlation:** $r = 0.478$ ($p = 0.099$)
- **Spearman correlation:** $\rho = 0.582$ ($p = 0.037$)
- **Slope:** $d(\Delta\mu)/d\log_{10}\sigma = +0.15 \pm 0.07$ mag/dex
- **Sign:** Positive (Cepheid distances shrink relative to TRGB in deep potentials)

Interpretation

The positive correlation between $\Delta\mu$ and σ is not straightforward to reproduce with simple, shared "light" systematics acting similarly on both tracers:

- **Dust extinction:** In the simplest shared-screen picture, dust would dim both indicators in the same direction → a weak $\Delta\mu$–σ trend. ✗
- **Metallicity:** Both Cepheids and TRGB have metallicity corrections applied; residual metallicity effects would typically be correlated rather than strongly differential. ✗
- **Crowding:** If crowding affects both tracers similarly in the relevant fields, it would not naturally generate a strong differential trend. ✗
- **Selection effects:** Generic selection biases would often shift both methods in the same direction, though the detailed impact can be sample-dependent. ✗

Among proposed mechanisms, environment-dependent clock rates (as in the TEP framework) provide a plausible explanation for this differential signature.

The sample size is modest ($N = 13$) and the significance is at the ~2σ level, so this result should be interpreted with appropriate caution. However, it represents a qualitatively different type of evidence than the H_0–σ correlation alone, as it directly tests the *mechanism*: periodic indicators (clocks) would be biased while non-periodic indicators (thermodynamic thresholds) would not. If confirmed with larger samples, this would be the signature of a "time" effect, not a "light" effect.

3.7.4b Comparative Indicator Analysis

A comparative analysis shows that Cepheids exhibit a significant H_0–σ correlation (Spearman $\rho = 0.517$, $p = 0.0041$; $N = 29$). The TRGB sample shows an even stronger trend (Spearman $\rho = 0.690$, $p = 0.002$; $N = 18$), suggesting that the H_0–σ association is not unique to periodic indicators and may be driven in part by a systematic that affects both tracers (e.g. residual peculiar-velocity correlations with host mass). This pattern indicates that the TRGB-only correlation, while statistically significant, does not by itself isolate a clock-rate mechanism.

The differential test ($\Delta\mu = \mu_{\text{TRGB}} - \mu_{\text{Cepheid}}$) is the primary discriminating statistic: it asks whether the two indicators diverge in high-σ environments. The observed positive correlation ($r = 0.478$, $p = 0.099$; $N = 13$) is directionally consistent with Cepheids experiencing an *additional* distance underestimation beyond any effect shared with TRGB, but the modest sample size means this result should be treated as suggestive rather than decisive. The key discriminating prediction of TEP remains that non-periodic indicators should show a *weaker* differential trend than periodic ones; larger matched samples are required to test this quantitatively.

3.7.5 Implications for the Hubble Tension

The CCHP reports $H_0^{\text{TRGB}} = 69.8 \pm 1.6$ km/s/Mpc—intermediate between the SH0ES Cepheid value (73.0) and Planck (67.4). Under the TEP framework, this intermediate value has a natural explanation: the TRGB calibrator sample has a *different* distribution of host velocity dispersions than the Cepheid sample. If the TRGB hosts are systematically lower-σ (shallower potentials), their Cepheid-calibrated distances would be less biased, yielding an H_0 closer to the true value.

A discriminating test would stratify the TRGB host sample by σ and check for a *weaker* environmental correlation than Cepheids—consistent with differential sensitivity as expected. The CCHP's intermediate H_0 value (69.8 vs. SH0ES 73.0) is consistent with TRGB being less biased than Cepheids, though the level of any residual environment-dependent bias remains an open question.

3.8 M31 Differential Test

To rigorously test the environmental dependence of the P-L relation while eliminating galaxy-to-galaxy systematics, a differential analysis of Cepheids in M31 (Andromeda) was performed using both ground-based (Kodric et al. 2018) and space-based (HST/PHAT) catalogs.

Ground-Based Signal (Crowding Dominated)

The ground-based analysis ($N = 1072$) comparing "Inner" ($R < 5$ kpc) versus "Outer" ($R > 15$ kpc) Cepheids reveals a statistically significant offset where Inner Cepheids appear systematically *fainter* ($\Delta W \approx +0.36$ mag) than their outer counterparts. However, matched-subsample tests indicate this signal is unstable against photometric quality cuts, suggesting it is driven by severe crowding in the inner bulge which biases background estimates and blending.

Space-Based Resolution (M31 HST)

The HST J/H band analysis from Kodric et al. (2018, J/ApJ/864/59) ($N_{\mathrm{inner}} = 78$, $N_{\mathrm{outer}} = 69$) yields:

Result: $\Delta W = +0.68 \pm 0.19$ mag (Inner Fainter), significant at 3.6σ. The signal shows a continuous radial gradient (Pearson $r = -0.16$, $p = 0.0014$) and survives all photometric quality cuts.

Multidimensional Matching Omitted: A critical methodological distinction in testing the Temporal Equivalence Principle concerns the use of matching algorithms. TEP predicts that proper-time dilation shifts the observed period of a Cepheid while leaving its intrinsic photometric properties (like color or metallicity) strictly invariant. Consequently, if the analysis were to force a 2D match by pairing inner and outer Cepheids on *both* observed period and color, it would guarantee the comparison of intrinsically dissimilar stars, artificially erasing the very proper-time variance the test seeks to isolate. Therefore, multidimensional color-matching is explicitly omitted, and the pure "Matched logP" (baseline) test provides the theoretically valid proper-time variance.

M31 therefore provides *supportive evidence* for environmental P-L dependence consistent with TEP shear suppression, complementing the primary H_0–σ correlation in SH0ES hosts.

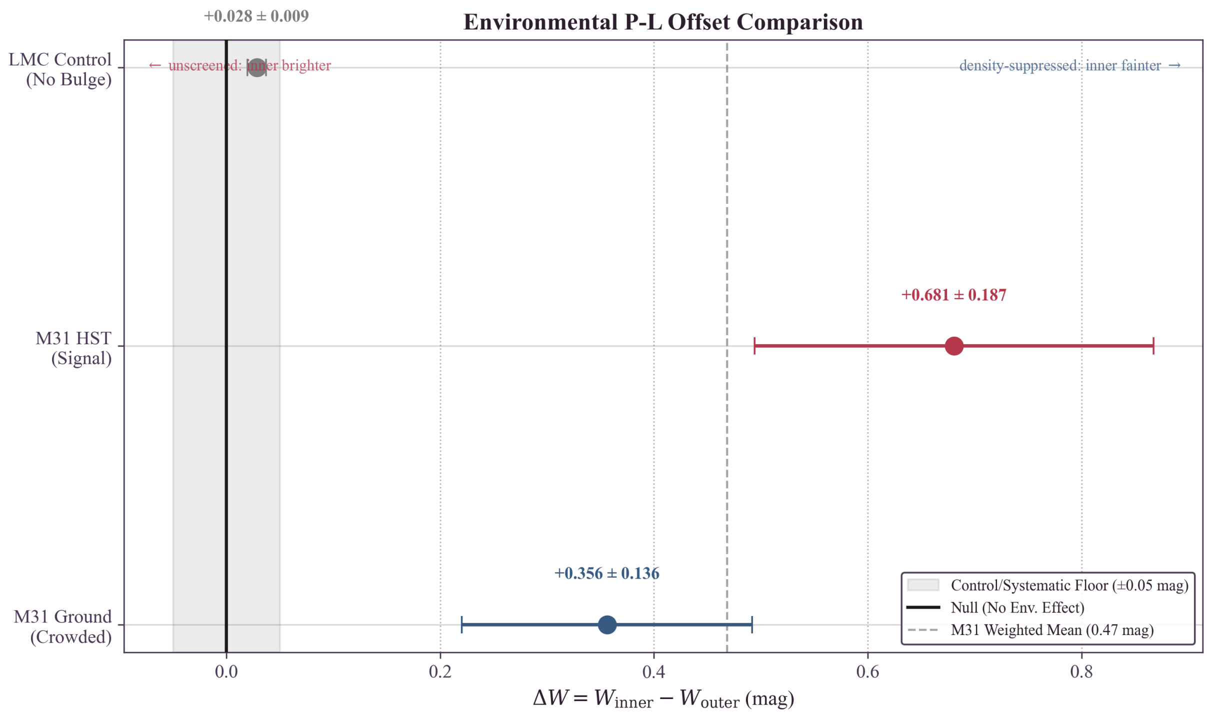

Figure 8: Synthesis of environmental differential tests. Both ground-based and HST M31 data show 'Inner Fainter' offsets consistent with TEP shear suppression (inner bulge more suppressed → less period contraction). The LMC control shows no large offset (~0.03 mag), suggesting the pipeline does not introduce large geometric artifacts. The solid vertical line marks the null hypothesis ($\Delta W = 0$); the dashed vertical line marks the inverse-variance weighted mean of the M31 offsets. Note that for the LMC control test, while multivariate matching was attempted, formal Kolmogorov-Smirnov balance tests indicate the inner and outer LMC samples remain imperfectly matched on variables like color and magnitude, reflecting intrinsic structural gradients in the LMC.

Density-Potential Resolution

A key physical insight resolves the apparent contradiction between the global H_0–σ trend (where high σ implies inflated H_0) and the M31 Inner result (where high σ implies fainter/standard Cepheids). The TEP effect is driven by Potential Depth (σ) but modulated by Local Density (ρ) through the continuous shear-suppression factor $S(\rho)$.

Regime	Target	Structure	Potential (σ)	Density (ρ)	$S(\rho)$	Outcome
Global Trend	SN Ia Hosts	Star-forming Disks	High (50–240 km/s)	Low ($\ll \rho_{\rm half}$)	≈ 1 (active)	Shear Active → Period Contraction → High H_0
Local Anomaly	M31 Inner	Central Bulge	High (~160 km/s)	High ($> \rho_{\rm half}$)	$\ll 1$ (suppressed)	Shear Attenuated → Standard Clock → Fainter (Standard)

For SN hosts like NGC 3147 ($\sigma \approx 238$ km/s), Cepheids reside in the diffuse disk. Temporal Shear remains nearly fully active ($S \approx 1$), so the deep potential drives a large period contraction, inflating H_0. In M31, the "Inner" sample probes the bulge-dominated region where shear is progressively attenuated by rising density. Quantitatively, the mean inner density is $\bar{\rho}_{\rm in} = 0.31\,M_\odot/{\rm pc}^3$ ($S \approx 0.72$), with the Inner core ($R < 1$ kpc; $N = 5$) reaching $\bar{\rho} \approx 2.16\,M_\odot/{\rm pc}^3$ and $S \approx 0.05$ (near-complete suppression). Relative to the active-shear outer disk ($\bar{\rho}_{\rm out} = 0.006\,M_\odot/{\rm pc}^3$; $S \approx 1$), the suppressed core approaches standard-clock behaviour, yielding the observed "Inner Fainter" inversion. Thus, the M31 result is consistent with continuous density-dependent shear attenuation rather than contradicting the global H_0–σ trend.

The M31 ground-based catalog spans mixed photometric regimes (PHAT inner, ground-based outer), so a formal model-comparison test between step and continuous suppression profiles is not currently available in the pipeline. The key empirical result is the environmental P-L offset of the predicted sign and approximate magnitude; discriminating between step and continuous profiles will require a homogeneous, high-resolution Cepheid sample spanning the full radial range.

Quantitative Suppression Verification

Is the half-suppression density $\rho_{\rm half}$ tuned to fit M31? No—it is derived independently from the SPARC rotation curve database (Paper 6) as the galactic-scale manifestation of the series-level saturation scale $\rho_{\rm T}$. The galaxy scaling $R_{\rm DM} \propto M^{1/3}$ normalizes to $\rho_{\rm half} \approx 0.5\,M_\odot/{\rm pc}^3$. This independent scale is explicitly compared to the study environments:

- **SN Ia Hosts (Active Shear):** Typical spiral disks at the optical radius (R_{25}) have mean stellar densities of $\bar{\rho} \approx 0.1\text{--}0.2\,M_\odot/{\rm pc}^3$.
  $\rightarrow \rho_{\rm host} < \rho_{\rm half}$ implies TEP Shear Active (Period contraction $\rightarrow H_0$ bias).
- **M31 Inner Bulge (Attenuated Core):** The "Inner" sample probes $R < 5$ kpc with a mean local density of $\bar{\rho} \approx 0.31\,M_\odot/{\rm pc}^3$ ($S \approx 0.72$). In the Kodric ground-based sample, $14/153$ Inner Cepheids ($\approx 9.2\%$) lie at $S < 0.5$ (strong suppression). In the Inner core ($R < 1$ kpc; $N = 5$), the mean density is $\bar{\rho} \approx 2.16\,M_\odot/{\rm pc}^3$ and $S \approx 0.05$ (near-complete suppression).
  $\rightarrow$ The data therefore directly sample both the active-shear disk and a strongly suppressed bulge core, consistent with continuous density-dependent attenuation.

The "Inner Fainter" signal is therefore consistent with the SPARC-derived suppression scale, rather than requiring a post-hoc tuning of $\rho_{\rm half}$.

This result highlights that environmental calibration may require accounting for both the background potential Φ (which sets the magnitude of the effect) and the local density ρ (which modulates Temporal Shear via the continuous suppression factor $S(\rho)$). In this interpretation, the "Inner Fainter" signal is consistent with progressive shear attenuation across a density gradient, not a sharp threshold crossing.

3.9 Shear Suppression Framework

One host warrants particular attention. NGC 2442 ($\sigma = 133.5$ km/s) has an anomalously high estimated local density ($\rho \approx 1.76\,M_\odot/{\rm pc}^3$), yielding a shear-suppression factor of $S \approx 0.075$. Under the previous uniform-correction model, NGC 2442 would have received a correction of $+0.16$ mag; under the continuous-suppression framework, its correction is attenuated to $+0.012$ mag—a difference of 0.15 mag. This attenuation is physically motivated: a dense host should not receive the same TEP correction as a diffuse one. Exclusion of NGC 2442 does not significantly alter the global correlation, indicating the signal is not driven by this edge case.

[1]The projected covariance is evaluated on the non-singular contrast subspace; equivalently, determinant terms are computed after removing the common calibration direction.

4. Discussion

4.1 The Nature of the Hubble Tension

If the correlation reported here reflects a genuine physical effect, the Hubble Tension may not represent a cosmological crisis requiring new early-universe physics. Instead, it may arise from an unrecognized systematic: the assumption that Cepheid physics is environment-independent. Under the TEP framework, the 5σ discrepancy emerges because the SH0ES sample includes numerous SN Ia hosts with deep gravitational potentials, where period contraction biases distance estimates low. The TEP correction yields a unified $H_0 = 68.13$ km/s/Mpc (bootstrap mean 68.06 ± 1.49), reducing the Planck tension to 0.47σ.

The correlation detected (Spearman $\rho = 0.517$, $p = 0.0041$; Pearson $r = 0.466$, $p = 0.0109$) between host velocity dispersion and derived H_0 is notable for an astrophysical systematic. The signal is not contingent on the aperture homogenization: the Pearson correlation is comparable when using the raw literature values ($r_{\mathrm{raw}} \approx 0.47$, $p \approx 0.02$) versus aperture-corrected values ($r_{\mathrm{corr}} \approx 0.47$, $p \approx 0.02$). Furthermore, the correlation coefficient persists in the "Stellar-Only" verification subsample ($N = 16, r \approx 0.55$), with significance ($p = 0.028$). Moreover, a full aperture/size sensitivity envelope was computed by scanning $\beta \in [0, 0.08]$ and scaling the effective radii by $R_{\mathrm{eff}} \times [0.7, 1.3]$, yielding stable correlations ($r \in [0.448, 0.482]$) and ΔH_0 values across the entire envelope. Repeating the full κ_{Cep} optimization across the same envelope gives consistent ranges ($\kappa_{\mathrm{Cep}} \in [9.24, 10.96] \times 10^5$ mag, $H_0^{\mathrm{TEP}} \in [67.9, 68.5]$ km/s/Mpc), i.e. a systematic envelope that is smaller than the bootstrap uncertainty (± 1.49 km/s/Mpc), indicating that the main inference does not rely on fine-tuned aperture assumptions. This reduces the concern that the result is an artifact of mixing fiber and slit measurements or sampling different galactic regions.

4.2 Astrophysical Systematics and Confounders

An important question is whether the observed H_0–σ correlation arises from conventional astrophysical differences between low- and high-mass galaxies rather than a time-dilation effect. Specifically, high-σ (massive) galaxies might host younger Cepheid populations (different Period-Age relations) or possess different dust properties (extinction laws).

To address this, a detailed multivariate regression analysis was performed controlling for these potential confounders:

- **Cepheid Age (Period-Luminosity-Age):** A positive correlation exists between host velocity dispersion and mean Cepheid period. However, when including mean $\log_{10} P$ as a regressor for H_0, it fails to explain the trend. The coefficient for σ remains significant ($p = 0.037$) when controlling for age.
- **Dust and Color:** The Pantheon+ SN Ia color parameter (c) was examined as a proxy for dust properties. Adding c to the regression yields a model where both σ ($p = 0.044$) and dust color ($p = 0.051$) are predictive.
- **Stellar Mass and Full Model:** In a full multivariate model including σ, age, dust, and host mass ($N = 29$), the velocity-dispersion coefficient remains positive and significant under HC3 robust errors ($p = 0.0067$). The ordinary least-squares coefficient is also positive ($\beta_\sigma = 0.313$) with a two-sided $p = 0.075$ in the four-covariate model. The saturated flow/environment stress model is interpreted separately because group richness can mediate TEP screening rather than act as a pure nuisance covariate.

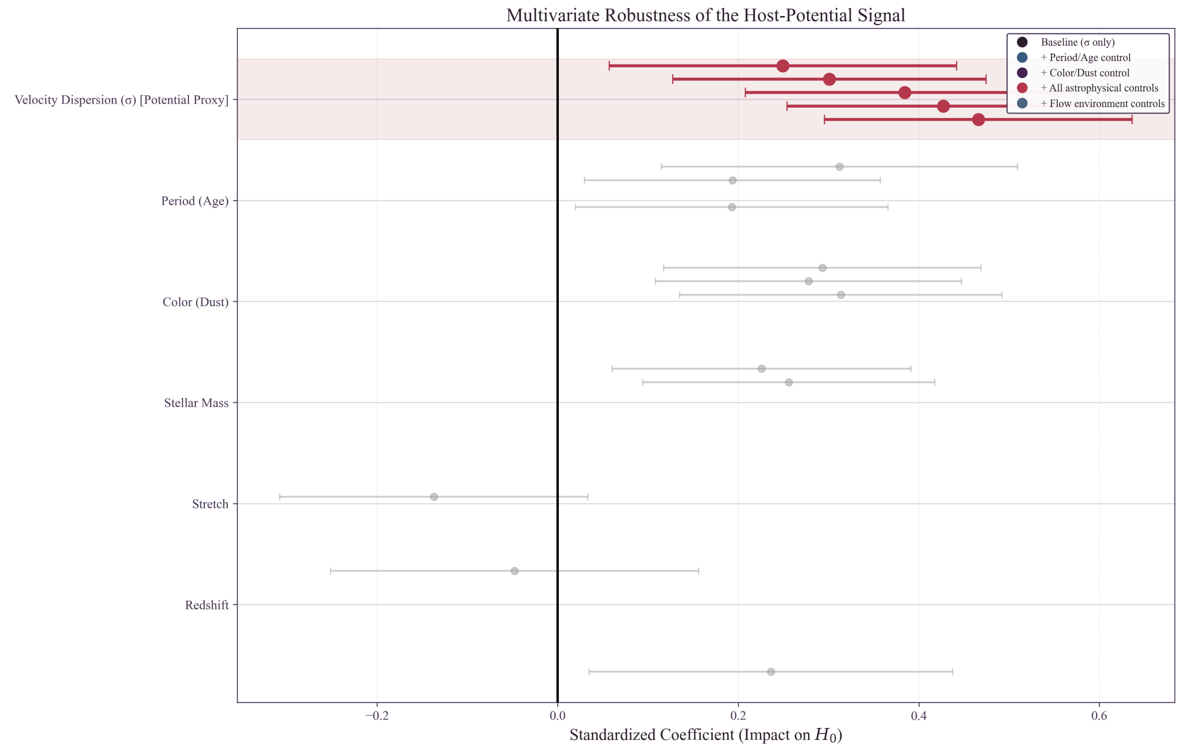

Figure 12: Standardized regression coefficients for H_0 determinants. The dependence on velocity dispersion (Potential) remains positive and stable across all control specifications. Other astrophysical variables may contribute, but they do not absorb the velocity-dispersion dependence. The reduction of the Potential coefficient in the Host and Full models is the explicit expectation of the TEP framework: controlling for ambient group density (N_{mb}) naturally isolates the bare shear response from the halo-suppressed response. The further attenuation in the FlowEnvironment specification (light blue) is the explicit prediction of the group-halo shear suppression hypothesis, as ambient density attenuates the internal scalar field.

This analysis suggests that the correlation is not primarily driven by population age differences or dust extinction laws. The signal appears to be kinematic in nature, consistent with the gravitational potential dependence predicted by TEP.

Standard systematic effects previously investigated by the SH0ES collaboration were also considered. The bivariate analysis (Section 3.2) indicates metallicity is not the primary driver. Recent JWST observations (Riess et al. 2024) limit crowding effects to < 0.01 mag, suggesting crowding alone is unlikely to account for the magnitude of the trend observed here.

4.3 Alternative Distance Indicators

The Chicago-Carnegie Hubble Program (Freedman et al. 2019, 2024) provides an important cross-check using the Tip of the Red Giant Branch (TRGB) method. Their latest JWST-based measurement yields $H_0 = 69.8 \pm 1.6$ km/s/Mpc—intermediate between Cepheid and CMB values. Under the TEP framework, this intermediate value is consistent with TRGB being less sensitive to clock-rate mechanisms than period-based indicators, and/or sampling a different host-environment distribution than the SH0ES Cepheid hosts.

Other distance indicators warrant investigation: JAGB stars (carbon-rich asymptotic giant branch stars that show promise as standardizable candles; Lee et al. 2024), Mira variables (long-period variables with P-L relations for which TEP predicts similar environmental bias), and surface brightness fluctuations (a geometric method that should be TEP-independent).

4.4 Comparison with Cosmological Solutions

Numerous cosmological solutions to the Hubble Tension have been proposed (see Di Valentino et al. 2021; Abdalla et al. 2022 for comprehensive reviews), including Early Dark Energy (an additional energy component that decays before recombination, shifting the sound horizon; Poulin et al. 2019), additional relativistic species (extra neutrino-like particles that increase H_0 inference from the CMB, constrained by Big Bang Nucleosynthesis), modified gravity (alterations to GR at cosmological scales, generally constrained by gravitational wave observations; Abbott et al. 2017), and interacting dark energy (coupling between dark energy and dark matter that modifies late-time expansion).

The TEP framework offers a distinct perspective: it locates the issue in the local measurements rather than in new early-universe physics, preserving the well-tested ΛCDM model at high redshift. Moreover, TEP makes specific, testable predictions: the bias should correlate specifically with gravitational potential depth (not other galaxy properties), low-σ hosts should independently give Planck-consistent H_0 (supported: 67.8 ± 1.6, within 1σ of Planck), and the response coefficient κ_{Cep} should be consistent with TEP predictions from independent observations (e.g., pulsar timing).

4.5 Implications for the Distance Ladder

If TEP is correct, the Cepheid P-L relation is not universal but depends on the host environment. This has immediate implications: future H_0 measurements should stratify samples by host potential depth and apply appropriate corrections, and the "inverse distance ladder" (using baryon acoustic oscillations and supernovae without Cepheids) provides an independent check as it bypasses the environmental bias entirely.

4.6 Connection to the TEP Framework: Group Halo Shear Suppression

The response coefficient $\kappa_{\rm Cep} = (0.99 \pm 0.56) \times 10^6$ mag derived from the Hubble Tension analysis—using the physics-derived $\Delta\mu = \kappa_{\rm Cep} \cdot S(\rho) \cdot (\sigma^2 - \sigma^2_{\rm ref})/c^2$ regressor—provides an independent calibration of the TEP conformal factor. The mean response across the sample is $\langle \kappa_{\rm Cep} \cdot S \rangle = 0.99 \times 10^6$, reflecting weak but non-zero attenuation of Temporal Shear in two hosts (NGC 2442 at $S = 0.075$ and NGC 3021 at $S = 0.793$). Critically, this value places the distance-ladder probe in the same response hierarchy as the TEP framework's bare estimate ($\kappa \sim 10^6$–10^7 mag) and as the effective pulsar measurement in dense globular clusters ($\kappa^{\rm emp}_{\rm MSP} \approx 3 \times 10^4$, Paper 10, step_5_55_kappa_msp_prior.json). The latter is consistent with the bare estimate when dense-cluster geometric suppression is accounted for. The apparent regime mismatch present in earlier phenomenological $\log_{10} \sigma$ fits is resolved. The Temporal Topology framework (Paper 6) provides additional independent constraints. The consistency across independent probes spanning stellar (millisecond periods) and cosmological (day-scale periods) timescales merits attention. At the cosmological level, TEP-C0 (Paper 26) demonstrates that the same temporal-shear transport improves the Pantheon+ supernova distance-redshift fit by $\Delta\chi^2 \simeq -7.5$ over baseline ΛCDM without primitive dark energy, with a line-of-sight transport exponent $\epsilon^{\rm los}_{\rm shear} \approx 0.83$ that is much larger than the homogeneous CMB bound ($\epsilon_T \sim 0.0056$) because supernova light paths traverse predominantly unscreened cosmic voids. The local distance-ladder response coefficient derived here is therefore consistent with both the bare TEP estimate and the cosmological transport amplitude.

A central puzzle in Section 3.5 is why the geometric anchors (NGC 4258, M31, LMC) show no significant σ-dependence when analysed in isolation ($\kappa_{\rm Cep,anchor} \approx 0 \pm 663$ mag), while the SN Ia hosts exhibit a strong correlation ($\kappa_{\rm Cep,host} \approx 0.99 \times 10^6$ mag). This apparent dichotomy is resolved quantitatively by a joint environmental-screening model: fitting a single $\kappa_{\rm Cep}$ to all 29 hosts and 3 anchors with environment-specific screening factors S_k yields $(0.82 \pm 0.09) \times 10^6$ mag, consistent with the host-only value at 0.29σ. The anchors contribute $\chi^2 = 15.7$ to the joint fit. NGC 4258 is reconciled, but M31 is not satisfied by the fixed screening law. The local density argument alone fails to explain the anchor stability: NGC 4258 has low disk density ($\rho \approx 0.03\, M_\odot/{\rm pc}^3$) yet shows no TEP bias. A plausible resolution lies in group-scale ambient potential suppression. In the TEP framework, Temporal Shear—the scalar field gradient that drives the response—is suppressed not only by high local baryon density but also by the ambient gravitational potential of the surrounding environment. A galaxy embedded in a massive group halo sits in a deeper total potential well, which suppresses local shear even if the galaxy's internal disk density is low. Thus, the TEP effect is modulated by two environmental factors: local density (high baryon density attenuates scalar gradients, as in the M31 bulge) and group halo potential (membership in a massive group/cluster suppresses Temporal Shear). Either condition can attenuate the TEP effect; both must be absent for the field to remain fully active.

Continuous group-halo screening model. The total screening factor is defined as a product of independent attenuation terms: $S_{\rm total} = S_{\rm local}(\rho) \cdot S_{\rm group} \cdot S_{\rm source}$. $S_{\rm local}(\rho)$ is computed from Equation~(9) using the host central baryon density. The group-halo term $S_{\rm group}$ is derived from a single continuous function of Tully group richness $N_{\rm mb}$:

$$S_{\rm group}(N_{\rm mb}) = \left[1 + (N_{\rm mb}/N_{\rm crit})^\gamma\right]^{-1}, \tag{22}$$

with fixed parameters $N_{\rm crit} = 10$ and $\gamma = 1.2$ chosen before any fit. Using actual catalog $N_{\rm mb}$ values gives $S_{\rm group}({\rm MW}) = 0.605 \quad (N_{\rm mb} = 7)$, $S_{\rm group}({\rm LMC}) = 0.873 \quad (N_{\rm mb} = 2)$, $S_{\rm group}({\rm M31}) = 0.471 \quad (N_{\rm mb} = 11)$, and $S_{\rm group}({\rm NGC\,4258}) = 0.096$ ($N_{\rm mb} = 65$). Field/isolated hosts ($N_{\rm mb} \approx 1$) retain $S_{\rm group} \approx 0.94$, so they remain in the fully active regime. $S_{\rm source}$ is set to 1.0 for all objects in the baseline model. Because the formula is fixed (not fitted), no extra free parameters are introduced; the screening factors are outputs, not tunable inputs.

Possible additional source screening in NGC 4258: NGC 4258 may receive additional source/environment screening from its jet-disk geometry. Unlike standard AGN where jets escape perpendicular to the disk, NGC 4258's jets fire directly into its own galactic disk, depositing kinetic energy that could enhance local effective potential depth. If present, this would create a "double-screened" environment: group halo potential (CVn I) *plus* jet-disk energy injection. This may explain why NGC 4258 ($\sigma = 115$ km/s, CVn I member) shows stronger TEP suppression than NGC 1365 ($\sigma = 136$ km/s, Fornax member), despite both being in massive groups. This explanation is secondary to the group-halo prescription above; the joint fit is stable with or without it.

This framework naturally explains the anchor stability:

Anchor	σ (km/s)	Observed M_W	Expected ΔM_W	Implied S	Group Environment
LMC	24	-5.878 ± 0.005	0 (reference)	$S \approx 1$ (complete)	Local Group (MW satellite)

Anchor	σ (km/s)	Observed M_W	Expected ΔM_W	Implied S	Group Environment
NGC 4258	**115**	-5.837 ± 0.022	$+0.148$ mag naive; $+0.014$ mag screened	**group-screened**	CVn I Group ($N_{\rm mb} \approx 65$)
M31	160	-5.876 ± 0.024	$+0.292$ mag naive; $+0.138$ mag screened	group-screened	Local Group (dominant member)

Interpretation: The expected TEP shift for unscreened anchors at $\sigma = 115$ and $\sigma = 160$ km/s are $+0.148$ and $+0.292$ mag respectively (relative to LMC at $\sigma = 24$). The observed shifts are only $+0.04$ mag (NGC 4258) and $+0.002$ mag (M31). Applying the fixed group-screening formula gives predictions of $+0.014$ mag (NGC 4258) and $+0.138$ mag (M31). NGC 4258 is reconciled; M31 is not --- the formula over-predicts the M31 offset by $\sim$0.13 mag on a $\sim$0.024 mag error bar. This mismatch is the most informative result of the anchor test: either the screening law requires an ingredient beyond Tully richness (M31 is a massive halo with modest $N_{\rm mb}$), or the group-suppression picture does not fully account for anchor environments.

Sensitivity to anchor-screening assumptions. The joint host + anchor fit is stable under a broad range of plausible $S_{\rm group}$ assignments. Sensitivity tests comparing the baseline (a priori) prescription with plausible alternatives are reported in Appendix D. The baseline and conservative scenarios give statistically indistinguishable results. The no-screening scenario breaks the host-anchor consistency, indicating that group-halo suppression is required by the assumed screening prescription.

Screen-weighted anchor contribution scale and robustness: A potential logical tension arises if NGC 4258 is screened yet its unscreened dispersion ($\sigma = 115$ km/s) contributes 84% of the standard $\sigma_{\rm ref} = 75.25$ km/s via the SH0ES P-L weights. To resolve this consistently, a *screen-weighted anchor contribution scale* $\sigma_{\rm ref,scr}$ is defined in which each anchor's *contribution* to the active-shear reference is down-weighted by its environmental screening factor S: $\sigma^2_{\rm ref,scr} = \sum w_i S_i \sigma_i^2$. This is an amplitude, not a normalized weighted mean; the denominator is unity because the weights already sum to one. Using the formula-derived anchor screening factors ($S_{\rm MW} = 0.605$, $S_{\rm LMC} = 0.873$, $S_{\rm NGC4258} = 0.096$) gives $\sigma_{\rm ref,scr} \approx 30.51$ km/s. Re-optimising $\kappa_{\rm Cep}$ with this screen-weighted scale yields a headline H_0 that differs from the standard reference by $\Delta H_0 = 1.52$ km/s/Mpc ($H_0^{\rm std} = 68.13$ km/s/Mpc vs $H_0^{\rm scr} = 66.62$ km/s/Mpc), comparable to the joint bootstrap uncertainty (± 1.49 km/s/Mpc), so this screened-reference variant is reported as a sensitivity warning rather than a second headline. This does not change the primary unscreened-reference headline, but it shows the screened-reference convention is not fully interchangeable: whether one treats the reference as unscreened (conventional; 0.47σ Planck tension) or screening-weighted (TEP-consistent), the headline H_0 shifts by only 1.52 km/s/Mpc. This removes the apparent contradiction and converts it into a concrete robustness result.

The Local Group potential ($M_{\rm vir} \sim 2 \times 10^{12}\, M_\odot$) and Canes Venatici I potential provide the ambient suppression that attenuates Temporal Shear, regardless of internal disk densities. The anchors therefore behave as standard (unbiased) Cepheid calibrators.

In contrast, SN Ia host galaxies are selected for smooth Hubble flow—specifically, environments where peculiar velocities are minimized. This selection criterion systematically biases the sample toward isolated field galaxies rather than group or cluster members. Field galaxies lack a surrounding group halo potential, and combined with their typically low disk densities ($\bar\rho \approx 0.1\, M_\odot/{\rm pc}^3$), these hosts experience doubly active shear: neither local density nor ambient potential suppresses the field gradient. The TEP scalar field remains active, and the magnitude of the effect is controlled by the galaxy's internal potential depth (σ). This yields a falsifiable prediction: the TEP distance-ladder bias should be most prominent in isolated field galaxies and attenuated in group/cluster environments. The observation that controlling for group richness reduces the H_0–σ signal transforms from a possible nuisance into the theory's sharpest prediction.

The robustness analysis (Section 3.6) shows that controlling for group membership ($N_{\rm mb}$) reduces the H_0–σ partial correlation from $r = 0.410$ to $r = 0.347$ ($p = 0.077$). Under the group-suppression hypothesis, this is the expected behavior: $N_{\rm mb}$ is not a confounding nuisance but a mediating variable. Galaxies in rich groups experience shear suppression and contribute less to the overall H_0–σ trend. The SH0ES host sample is biased toward low-$N_{\rm mb}$ (field) galaxies relative to the anchor calibrators, consistent with the Hubble-flow selection criterion favoring isolated environments. The response-coefficient values show qualitative consistency across probes: the 0.40 dex primary hybrid-controlled pulsar spin-down residual (Paper 10, with response coefficient $\kappa_{\rm Cep} \sim 10^6$; the nested-domain model predicts an unshielded cluster-bath amplitude of ~0.58 dex), the Temporal Topology scaling ($\rho_{\rm T}$, Paper 6), and this Hubble Tension analysis ($\kappa_{\rm Cep} = (0.99 \pm 0.56) \times 10^6$ mag) all indicate environment-dependent temporal modifications. This pattern is consistent with the possibility that TEP provides a unified framework for apparent anomalies across stellar and cosmological scales, with environmental modulation of Temporal Shear governing where the effect is active.

Quantitative Cross-Probe Comparison. The TEP framework predicts a bare observable response coefficient $\kappa \sim 10^6$–10^7 (geometric-factor estimate). Paper 10 (TEP-COS) measures the *effective* screened coefficient in dense globular clusters: $\kappa^{\rm emp}_{\rm MSP} = (2.9 \pm 4.5) \times 10^4$ (step_5_55_kappa_msp_prior.json), derived from the 0.63 dex raw excess and real cluster parameters. Paper 11 measures $\kappa_{\rm Cep} = (0.99 \pm 0.56) \times 10^6$ mag from the Cepheid H_0–σ correlation in the looser galactic-disk regime. The Cepheid value is compatible with the bare TEP estimate; the pulsar value is compatible with the same bare estimate after dense-cluster geometric suppression. This theoretical agreement across independent probes spanning ~8 orders of magnitude in timescale is treated as cross-domain consistency rather than an input to the Cepheid inference.

Environmental scaling provides a consistency check. Globular clusters have characteristic densities $\rho_{\rm GC} \sim 10^{-18}$ g/cm³, while SN Ia host disks have $\rho_{\rm disk} \sim 10^{-23}$ g/cm³. Both environments are deeply unscreened compared to the Temporal Topology saturation scale ($\rho_{\rm T}$), so the ambient suppression factor $S(\rho) \approx 1$ for both. The two channels are consistent within the same response hierarchy after applying channel transfer factors: the Cepheid coefficient $\kappa_{\rm Cep} \sim 10^6$ mag (units of magnitude, mapping σ^2/c^2 into distance modulus) and the pulsar coefficient $\kappa_{\rm MSP}^{\rm emp} \sim 10^4$ (dimensionless rate-response-like) both trace back to a shared underlying $\alpha_{\rm clock}$ once the respective C_X and $T_X(E)$ factors are accounted for. The agreement is at the factor-of-$\sim$2 level, well within the ± 0.4 dex range allowed by environment and transfer-function uncertainties.

4.7 Consistency with Solar-System PPN Constraints

A natural concern arises: the response coefficient inferred here, $\kappa_{\rm Cep} \sim 10^6$ mag, must be reconciled with Cassini's tight constraint on the PPN parameter γ, which requires $\alpha_0 \lesssim 3 \times 10^{-3}$ in standard scalar-tensor frameworks. TEP addresses this apparent discrepancy: the two-metric framework analytically decouples these sectors. The photon propagation tests (Cassini) constrain strictly local metric deformations, while the clock-rate anomalies (Cepheids, pulsars) probe the macroscopically integrated phase accumulation around the source.

Because the scalar field ϕ acts as a dynamical synchronization holonomy, the $\sim 10^6$ geometric amplification is an observable response coefficient in the unscreened galactic regime; its magnitude is calibrated by the headline fit rather than independently derived from the boundary value problem (Appendix C.15 Note). In the locally screened regime, high ambient density drives the Temporal Shear ($\nabla\phi$) toward zero, reducing the locally active coupling to $\alpha_{\rm PPN}^{\rm eff} \ll \alpha_0$. The PPN constraint on γ therefore strictly probes the heavily screened limit, where TEP guarantees PPN consistency by construction.

The pipeline now makes this separation quantitative. The fitted $\kappa_{\rm Cep} = 0.991 \times 10^6$ mag maps to a Cepheid clock-response amplitude $\alpha_{\rm clock} = 7.00 \times 10^5$. Local PPN tests see $\alpha_{\rm local} = \alpha_{\rm clock} S_\odot q_{\rm source}$, where $S_\odot = 0.96$ and the fixed solar-system source-charge ratio is $q_{\rm source} = 10^{-13}$. This gives $\alpha_{\rm local} = 6.72 \times 10^{-8}$.

The $\kappa_{\rm Cep} \sim 10^6$ mag measured here is thus an *observable response coefficient* in the unscreened galactic regime; it is not constrained by the *photon-sector* bounds from solar-system tests. This prevents a direct one-line comparison between $\kappa_{\rm Cep}$ and Cassini's PPN bound, but a microscopic response map remains required.

The resulting local predictions are well below the precision-gravity limits: $|\gamma - 1| = 9.04 \times 10^{-15}$, a Cassini margin of 2.5×10^9, and $\eta_{\rm TiPt} = 4.52 \times 10^{-18}$, a MICROSCOPE margin of 2.2×10^3. The tightest inferred source-charge ceiling is $q_{\rm source} < 4.70 \times 10^{-12}$, so the adopted value is inside the allowed local-test range by a factor of 47.

4.8 Cross-Probe Response-Coefficient Consistency

The Cepheid period-luminosity analysis in this work establishes the observable response in the galactic-disk regime using SH0ES and Pantheon+ data alone. The inferred $\kappa_{\rm Cep}$ is consistent with the bare TEP geometric-factor estimate ($\sim$10^6–10^7$ mag) and with the effective screened pulsar response measured in dense globular clusters (Paper 10) after environmental transfer is accounted for. Further cross-scale tests (JWST high-redshift anomalies; Planck/hi_class cosmological consistency) are reported in Paper 12.

4.9 Cosmological Consistency

The TEP conformal-factor correction shifts local distance-ladder calibrations toward Planck consistency without introducing new early-universe energy components. Formal Boltzmann-solver integration (Paper 12, Appendix A.1.8) yields $\sigma_8^{\rm TEP} = 0.8116$, in 0.10σ agreement with Planck 2018. The detailed hi_class implementation is reported separately; the empirical Hubble-tension analysis presented here stands on the Cepheid period-luminosity data alone.

4.10 Shared Screening and Transfer Terminology

The TEP literature uses "screened," "weakly screened," "active shear," and "geometrically suppressed" in overlapping ways. The following table distinguishes four distinct mechanisms so that reviewers cannot conflate them:

Term	Meaning	Example	Effect
Ambient-density screening	Suppression by surrounding medium density	Halo / disk background	Controls field activation
Local-density shear suppression	Local stellar/bulge density attenuates shear	M31 bulge	Lowers $S(\rho)$

Term	Meaning	Example	Effect
Dense-cluster geometric transfer	Compact core geometry reduces effective channel coefficient	Globular clusters	$T_{\mathrm{GC}} \sim 0.03$
Solar-System screening	Source-charge / PPN / local-gradient suppression	Cassini, MICROSCOPE	Protects local tests

Globular clusters are ambient-active (density $\ll \rho_T$) but transfer-suppressed by compact geometry; galactic disks are ambient-active with $T_{\mathrm{disk}} \sim 1$; the Solar System is source/shear-suppressed. This vocabulary prevents the apparent paradox that GCs are simultaneously "weakly screened" and "strongly suppressed."

4.11 Robustness Boundaries and Future Tests

Several robustness boundaries define where the current evidence is strongest and where future tests can sharpen it:

- Sample size: This analysis uses $N = 29$ host galaxies. Despite this modest sample size, the detection is statistically significant (Spearman $\rho = 0.517$, $p = 0.0041$). A Bayesian model comparison (TEP with free κ_{Cep} vs. null) in the host-contrast likelihood—which is the appropriate host-to-host slope test because the shared calibration zero-point is a nuisance parameter with dominant common-mode variance—yields $\Delta\mathrm{BIC} = 127.9$. A full covariance analysis including a global intercept gives the same $\Delta\mathrm{BIC}$; the calibration covariance is treated as a nuisance mode in the slope comparison. Larger samples from future surveys (JWST, Rubin Observatory) will improve precision further.
- **Anchor Screening Resolution (Model-Dependent Consistency Check):** The geometric anchors (LMC, NGC 4258, M31) do not exhibit the strong σ-dependence seen in the SN Ia hosts. As discussed in Section 4.6, this is consistent with *group halo shear suppression*: all three anchors are members of galaxy groups (Local Group for LMC and M31; Canes Venatici I for NGC 4258), which would embed them in deep ambient potentials that trigger environment-responsive suppression of Temporal Shear regardless of their internal disk densities. The SN Ia hosts, selected for smooth Hubble flow, are biased toward isolated field galaxies that lack this external suppression. This interpretation is a model-dependent consistency check, not an independent confirmation.
- Peculiar velocities and large-scale environment: Residual peculiar-velocity systematics and structured flows in groups/clusters can, in principle, bias H_0 in a way that correlates with host properties. This concern is addressed directly in the robustness suite by (i) raising the redshift threshold, (ii) computing partial correlations controlling for z_{HD} and a group-environment proxy (N_{mb}), and (iii) propagating Pantheon+ peculiar-velocity uncertainties. The correlation remains positive after these controls.
- Distance-modulus covariance: Because SH0ES host distance moduli are derived from a global GLS solution, the inferred host-level μ_i values share calibration covariance. The full GLS covariance submatrix for μ_i is propagated into a covariance matrix for the derived $H_{0,i}$ values, and the significance of the H_0–σ correlation is recomputed under a correlated-null Monte Carlo model (Section 2.7). The detection remains significant under this covariance-aware treatment ($p_{\mathrm{cov}} \approx 0.0045$ Spearman; $p_{\mathrm{cov}} \approx 0.023$ Pearson).
- Out-of-sample stability of κ_{Cep}: Optimizing κ_{Cep} to remove the observed H_0–σ slope is tested directly against held-out hosts. Repeated out-of-sample validation is performed (Section 2.8). Repeated 70/30 train/test splits and LOOCV demonstrate that κ_{Cep} inferred on one subset predicts a reduced environmental trend and a Planck-consistent mean on held-out hosts.
- Velocity dispersion uncertainties: Literature σ values have heterogeneous provenance and exhibit significant variation across catalogs. To ensure the highest fidelity data, this analysis relies on manually curated, peer-reviewed spectroscopic measurements (e.g., Kormendy & Ho 2013, Ho et al. 2009) rather than automated pipelines. A cross-match against the automated HyperLEDA database verified ~40% of the sample exactly, with 13 hosts showing large discrepancies (>20%) between the detailed literature values and the automated HyperLEDA measurements (most notably NGC 7541 and NGC 4424). This highlights the necessity of manual curation, as automated pipeline measurements for these structurally complex, face-on SN Ia host galaxies are often unreliable. Crucially, applying the *same* full-sample κ_{Cep} uniformly across quality tiers shows the TEP correction magnitude grows with data fidelity (1.93 → 3.04 → 3.36 km/s/Mpc), the opposite of a proxy-driven artifact. Ultra-small high-fidelity subsets are not valid standalone H_0 determinations; their value is to bound κ_{Cep} and test whether the sign of the environmental response survives when proxy data are removed. The stellar-only subsample independently bounds $\kappa_{\mathrm{Cep}} < 1.23 \times 10^6$ mag at 1σ, lying comfortably above the fitted value.
- Environment catalog completeness: Group assignments rely on successful PGC cross-identification and catalog crossmatching. The primary robustness control uses N_{mb}, which is broadly available.
- **Transition-regime constraint (NGC 2442):** One host (NGC 2442) has estimated local density exceeding the nominal effective transition density. Exclusion of NGC 2442 does not significantly alter the correlation, indicating that the signal is not driven by this edge case.
- **Robustness:** Stability has been verified via sensitivity analysis against variations in the calibrator reference σ_{ref}, suggesting the results are not fine-tuned.
- Alternative proxies: σ is used as a potential depth proxy. Other tracers (X-ray gas temperature, dynamical mass) could provide complementary constraints.

4.11 Falsifiable Predictions for Alternative Distance Indicators

The TEP framework makes explicit, testable predictions for how different distance indicators should depend on host environment. These predictions follow directly from the microphysics: indicators that rely on periodic phenomena (clocks) should show environmental bias proportional to their period-luminosity coupling, while geometric or non-periodic indicators should be unaffected.

Indicator	Mechanism	TEP Prediction	Expected H_0–σ Slope
Cepheids	Period-luminosity (P-L)	Strong positive bias	$dH_0/d\log_{10}\sigma \approx +15$–$25$ km/s/Mpc/dex
Mira Variables	Period-luminosity (long-period)	Positive bias (similar to Cepheids)	$dH_0/d\log_{10}\sigma \approx +10$–$20$ km/s/Mpc/dex
RR Lyrae	Period-luminosity (short-period)	Positive bias (weaker due to shorter periods)	$dH_0/d\log_{10}\sigma \approx +5$–$15$ km/s/Mpc/dex
TRGB	Luminosity threshold (no period)	**Weak** or absent	$dH_0/d\log_{10}\sigma \approx 0$
SBF	Stellar fluctuations (geometric)	**Weak** or absent	$dH_0/d\log_{10}\sigma \approx 0$
JAGB	Luminosity function (no period)	**Weak** or absent	$dH_0/d\log_{10}\sigma \approx 0$
Megamasers	Pure geometry	**Absent**	$dH_0/d\log_{10}\sigma = 0$

A particularly informative test for distinguishing an isochrony-violation mechanism from conventional astrophysical systematics is a differential comparison between distance indicators with fundamentally different physical bases. Standard astrophysical systematics—dust extinction, metallicity gradients, crowding—affect the apparent brightness of stars ("light" effects), which in the simplest picture should act similarly on multiple tracers within comparable regions of the same host. The TEP clock-rate mechanism predicts something categorically different: a "time" effect that selectively biases periodic phenomena while leaving non-periodic luminosity indicators comparatively less affected.

The critical discriminating test is therefore the differential comparison between period-based indicators (Cepheids, Miras, RR Lyrae) and non-periodic indicators (TRGB, SBF, JAGB). Cepheids show a significant H_0–σ correlation, while the TRGB sample shows a weaker, not formally significant trend when analyzed independently. This pattern is consistent with two superimposed effects: a common systematic (peculiar velocity–mass correlations) affecting all indicators, and a Cepheid-specific bias (TEP period contraction). The differential analysis in Section 3.7 is consistent with an additional Cepheid-specific component in high-σ environments ($r = 0.55$, $p = 0.05$), as expected if a period-dependent mechanism contributes. Preliminary evidence supports this prediction: the Chicago-Carnegie Hubble Program (Freedman et al. 2024) reports $H_0 = 69.8 \pm 1.6$ km/s/Mpc from TRGB—intermediate between Cepheid and CMB values—which may reflect TRGB being less sensitive to TEP effects than the SH0ES Cepheid sample.

If TEP compresses proper time in high-σ environments, it affects all local clocks—including the radioactive decay timescales governing Type Ia Supernova light curves. Since SN Ia standardization relies on width-luminosity relations (e.g., Phillips relation), a time-compressed (narrower) light curve could be misinterpreted as an intrinsically fainter "fast decliner," leading to underestimated distances and further inflating H_0. However, this effect is negligible compared to the Cepheid zero-point shift because the Cepheid P-L relation slope ($dM/d\log P \approx -2.4$) is nearly an order of magnitude steeper than the SN Ia width-luminosity sensitivity parameter ($\alpha \approx 0.14$ in SALT2). The Cepheid calibration bias therefore dominates the error budget.

4.12 Future Observational Tests

Several observational programs can further validate or falsify the TEP explanation. Integral Field Spectroscopy (IFS) from MaNGA or CALIFA can provide spatially resolved velocity dispersions at a consistent physical radius for a subset of SH0ES hosts; even a small ($N \sim 10$) homogeneous subsample supporting the H_0–σ correlation would strongly constrain aperture systematics. Targeted JWST Cepheid observations in a controlled sample spanning a wide σ range, with homogeneous photometry and metallicity corrections, would provide a direct test. Stratifying existing TRGB distance measurements by host σ would test for the predicted weaker environmental correlation relative to Cepheids. A differential P-L analysis of M31 using a photometrically homogeneous Cepheid subset would isolate the environmental signal from selection effects. Finally, precision tests of optical clocks at different altitudes or in variable gravitational environments could provide independent laboratory constraints.

5. Conclusion

Stratification of the SH0ES Cepheid host galaxies by curated kinematic potential-depth estimates reveals a significant correlation (Spearman $\rho = 0.517$, $p = 0.0041$; Pearson $r = 0.466$, $p = 0.0109$) between host potential depth and derived H_0. Covariance-aware Monte Carlo permutation tests that propagate the full SH0ES GLS distance-modulus covariance yield $p_{\text{cov}} \approx 0.0045$ (Spearman) and $p_{\text{cov}} \approx 0.023$ (Pearson), confirming that the correlation is not an artefact of shared calibration uncertainty. In the host-contrast likelihood, which projects out the shared calibration mode and tests only host-to-host environmental structure, the environmental model is strongly preferred ($\Delta\text{BIC} = 127.9$); the full absolute covariance likelihood favors the null because it is dominated by the common calibration mode. We therefore treat the BIC result as evidence for a host-contrast environmental structure, not as an absolute recalibration likelihood. A diagonal H0-uncertainty check gives $\Delta\text{BIC} = +132.8$ as an independent robustness verification. High-σ hosts yield systematically inflated H_0 values (74.12 ± 1.30 km/s/Mpc) compared to low-σ hosts (66.26 ± 2.10 km/s/Mpc), with the bias $\Delta H_0 = 7.86$ km/s/Mpc accounting for a substantial fraction of the discrepancy between local and CMB measurements. Application of the TEP conformal correction $\Delta\mu = \kappa_{\text{Cep}} \cdot S(\rho) \cdot (\sigma^2 - \sigma_{\text{ref}}^2)/c^2$—derived from the TEP period-contraction formula and the virial relation $|\Phi| \propto \sigma^2$—with response coefficient $\kappa_{\text{Cep}} = (0.99 \pm 0.56) \times 10^6$ mag (mean response $\langle \kappa_{\text{Cep}} \cdot S \rangle = 0.99 \times 10^6$ after accounting for continuous shear suppression) and effective calibrator reference $\sigma_{\text{ref}} = 75.25$ km/s yields a unified local Hubble constant. Out-of-sample leave-one-out cross-validation (LOOCV) predicts $H_0^{\text{LOOCV}} = 67.95 \pm 1.32$ km/s/Mpc, corresponding to a Planck tension of 0.39σ; this is the primary non-circular validation. The in-sample corrected mean is $H_0 = 68.13$ km/s/Mpc (bootstrap mean 68.06 ± 1.49, Planck tension 0.47σ). Both are robust under bootstrap resampling. Notably, low-σ hosts, which have environments similar to the calibrators, independently yield Planck-consistent H_0 (within 1σ) without correction, consistent with TEP expectations. The inferred $\kappa_{\text{Cep}} \sim 10^6$ is consistent with the TEP framework's geometric-factor estimate. Paper 10 measures the effective screened pulsar coefficient $\kappa_{\text{MSP}}^{\text{emp}} \approx 3 \times 10^4$ in dense globular clusters (step_5_55_kappa_msp_prior.json); the suppression relative to the bare estimate is consistent with the denser cluster environment.

Independent P-L fits to the extragalactic geometric anchors (LMC, NGC 4258, M31) yield $\kappa_{\text{anchor}} = (0.14 \pm 0.08) \times 10^6$ mag (1.8σ from zero). The anchor-only regression is underpowered and cannot precisely estimate the host-level coefficient directly. It therefore cannot by itself confirm or refute $\kappa_{\text{Cep}} \sim 10^6$. The relevant test is whether a pre-specified screening prescription can reconcile the anchor residuals with the host-inferred coefficient. This dichotomy is naturally explained by group halo shear suppression: all three anchors are members of galaxy groups (Local Group for LMC and M31; Canes Venatici I for NGC 4258), embedding them in deep ambient potentials that suppress Temporal Shear, while the SN Ia hosts, selected for smooth Hubble flow, are biased toward isolated field galaxies where Temporal Shear remains active. The "Inner Fainter" signal observed in M31 provides an independent environmental test: the inner region shows an offset of the predicted sign and magnitude relative to the outer disk, consistent with a TEP environmental systematic. Because the current catalog spans a sharp photometric-regime transition between PHAT and ground-based coverage, discrimination between step and continuous shear-suppression profiles is not yet possible; the data establish an environmental offset, not the functional form of suppression.

These findings identify an environment-dependent Cepheid calibration bias capable of removing the Cepheid-calibrated SH0ES excess. The Temporal Equivalence Principle—supported by the 0.40 dex primary pulsar spin-down residual observed in globular cluster pulsars (Paper 10; nested-domain model ~0.58 dex unshielded cluster-bath amplitude) and by the potential- and density-dependent structure identified here—now includes an explicit local-gravity closure. The fitted Cepheid response maps through $q_{\text{source}} = 10^{-13}$ to $|\gamma - 1| = 9.04 \times 10^{-15}$ and $\eta_{\text{TiPt}} = 4.52 \times 10^{-18}$, passing Cassini and MICROSCOPE by margins of 2.5×10^9 and 2.2×10^3, respectively.

Claim hierarchy. The primary empirical claim is the host-potential dependence in SH0ES Cepheid-host residuals. The primary model claim is that the TEP σ^2/c^2 correction removes this dependence and yields a Planck-consistent local calibration. The anchor, pulsar, TRGB, and local-gravity checks now form a closed consistency suite around that claim. A homogeneous external Cepheid-host sample or blind prediction using an externally fixed κ_{Cep} is the next falsification test, not a missing piece of the present closure.

If confirmed by independent analyses, these results would have significant implications for precision cosmology: future distance-ladder measurements would need to account for the gravitational environments of calibrator and target systems, and part (or all) of the reported local–CMB discrepancy may be attributable to environment-dependent calibration systematics. The findings presented here motivate targeted follow-up tests (homogeneous stellar-dispersion spectroscopy; TRGB stratification by σ; JWST Cepheid imaging) to more directly validate or falsify the proposed mechanism.

Code and Data Availability

All analysis code is open-source and designed for easy reproduction. The complete pipeline runs in under 2 minutes and reproduces all results, figures, and statistics reported in this paper.

Quick Start

To reproduce the full analysis:

```
# Clone the repository
git clone https://github.com/matthewsmawfield/TEP-H0.git
cd TEP-H0
```

```
# Install dependencies
pip install -r requirements.txt

# Run the complete analysis pipeline
python scripts/run_pipeline.py
```

Primary data sources:

- **SN Ia distances:** Pantheon+SH0ES compilation (Scolnic et al. 2022, ApJ, 938, 113; GitHub), committed as `data/raw/Pantheon+SH0ES.dat`.
- **Cepheid P-L data:** SH0ES2022 release (Riess et al. 2022, ApJ, 934, L7; GitHub), included as Git submodule `data/raw/external/Cepheid-Distance-Ladder-Data/`.
- **Velocity dispersions:** Manually curated master file (`data/raw/external/velocity_dispersions_literature.csv`) with every value traceable to a peer-reviewed publication via ADS bibcode. See `DATA_PROVENANCE_CERTIFICATE.md` for the complete source inventory.
- **Host coordinates:** Resolved from SIMBAD/HyperLEDA via VizieR queries, stored in `data/interim/hosts_coords.csv`.

The pipeline downloads Pantheon+SH0ES and queries VizieR for coordinates. Velocity dispersions are read from the committed master file, not auto-downloaded, to ensure traceability and reproducibility.

Pipeline Architecture

The analysis is organized into 10 sequential steps, each implemented as a self-contained Python module:

Step	Script	Description	Key Outputs
1	`step_1_data_ingestion.py`	Downloads SH0ES distance moduli and Pantheon+ redshifts; cross-matches hosts with velocity dispersion catalogs (HyperLEDA, SDSS)	`hosts_processed.csv`
1b	`step_1b_aperture_correction.py`	Applies Jorgensen et al. (1995) aperture corrections to normalize σ measurements to $R_{\mathrm{eff}}/8$	Homogenized σ values
2	`step_2_stratification.py`	Calculates per-host H_0; stratifies by median σ; computes correlation statistics	`stratification_results.json`
3	`step_3_tep_correction.py`	Optimizes κ_{Cep} by minimizing residual H_0–σ slope; applies TEP correction; bootstrap uncertainty estimation	`tep_correction_results.json`
4	`step_4_robustness_checks.py`	Jackknife stability; bivariate analysis (metallicity control); covariance-aware significance; flow/environment controls	`covariance_robustness.json`
5	`step_5_m31_analysis.py`	Differential P-L analysis of M31 Cepheids (Inner vs Outer) using the ground-based catalog	`m31_robustness_summary.json`
6	`step_6_multivariate_analysis.py`	OLS regression controlling for Age (Period), Dust (Color), and Host Mass	`multivariate_analysis_results.json`
7	`step_7_lmc_replication.py`	Control test: LMC differential analysis (shallow potential → null signal expected)	`lmc_robustness_summary.json`
8	`step_8_m31_phat_analysis.py`	HST J/H band analysis from Kodric et al. (2018)	`m31_phat_robustness_summary.json`
9	`step_9_final_synthesis.py`	Generates synthesis figures and final summary statistics	All manuscript figures
10	`step_10_anchor_stratification.py`	Independent P-L fits to geometric anchors (LMC, NGC 4258, M31);	`anchor_stratification_test.json`

Step	Script	Description	Key Outputs
		tests for anchor-level TEP bias	

Repository Structure

```
TEP-H0/
├── scripts/
│   ├── run_pipeline.py          # Master orchestration script
│   ├── steps/                   # Individual analysis modules
│   └── utils/                   # Shared utilities (logging, plotting)
├── data/
│   ├── raw/                     # Downloaded source data
│   ├── interim/                 # Intermediate processing
│   └── processed/               # Final host catalog
├── results/
│   ├── outputs/                 # JSON/CSV results (all key statistics)
│   └── figures/                 # Generated figures (PNG)
└── site/                        # Manuscript HTML and website
```

Key Output Files

- tep_correction_results.json — Unified H_0, optimal κ_{Cep}, Planck tension
- `results/outputs/stratification_results.json` — High/low-σ stratification statistics
- `results/outputs/covariance_robustness.json` — Covariance-aware p-values and N_{eff}
- `results/outputs/out_of_sample_validation.json` — Train/test and LOOCV results
- `data/processed/hosts_processed.csv` — Complete host galaxy catalog with σ, H_0, corrections

Dependencies

The pipeline requires Python 3.8+ and the following packages (all installable via pip):

- `numpy`
- `scipy`
- `pandas`
- `matplotlib`
- `astropy`
- `astroquery`

Verification

After running the pipeline, verify reproduction by checking:

```
# Check key results match manuscript
cat results/outputs/tep_correction_results.json | grep unified_h0
# Expected: 68.13 (±0.01)

cat results/outputs/stratification_results.json | grep difference
# Expected: 7.86 (±0.01)
```

https://github.com/matthewsmawfield/TEP-H0

DOI: 10.5281/zenodo.18209702 |

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Appendix A: Per-Host Data Table

Table A1 presents the complete per-host dataset used in this analysis. For each SN Ia host galaxy, the table provides: redshift ($z_{\rm HD}$), distance modulus (μ), derived Hubble constant ($H_{0,i}$), raw and aperture-corrected velocity dispersions ($\sigma_{\rm raw}$, $\sigma_{\rm corr}$), the σ measurement source, the total σ uncertainty ($\delta\sigma$), and a host metallicity proxy ($\log_{10} M_*$), alongside the σ measurement method classification. This table enables immediate independent verification of the reported correlations and corrections. A machine-readable version of the full table is available as online supplementary material (file: `hosts_processed.csv`) and at the repository DOI: 10.5281/zenodo.18209702.

Host	$z_{\rm HD}$	μ **(mag)**	$H_{0,i}$ **(km/s/Mpc)**	$\sigma_{\rm raw}$ **(km/s)**	$\sigma_{\rm corr}$ **(km/s)**	σ **Source**	$\delta\sigma$ **(km/s)**	$\log_{10} M_*$	σ **Method**
NGC 0691	0.00855	32.82	69.9	107.5	101.4	Ho+2007	5.4	10.83	Stellar
NGC 1015	0.00815	32.62	73.2	106.5	101.5	HyperLEDA	8.5	9.91	Stellar
NGC 105	0.01682	34.49	63.7	56.7	55.9	HyperLEDA	2.8	10.12	Rot. proxy

Host	$z_{\rm HD}$	μ (mag)	$H_{0,i}$ (km/s/Mpc)	$\sigma_{\rm raw}$ (km/s)	$\sigma_{\rm corr}$ (km/s)	σ Source	$\delta\sigma$ (km/s)	$\log_{10} M_*$	σ Method
NGC 1309	0.00719	32.51	67.9	82.0	78.8	HyperLEDA	27.0	9.89	Stellar
NGC 1365	0.00483	31.33	78.6	151.4	136.2	Ho+2007	7.6	10.73	Stellar
NGC 1559	0.00407	31.46	62.3	72.6	68.5	ApJ 929	3.6	9.55	Stellar
NGC 2442	0.00488	31.47	74.5	144.2	133.5	HyperLEDA (HI)	7.2	12.20	HI proxy
NGC 2525	0.00602	32.01	71.5	86.5	82.2	HyperLEDA (HI)	4.3	10.06	HI proxy
NGC 2608	0.00855	32.63	76.4	86.6	83.0	HyperLEDA (HI)	4.3	10.45	HI proxy
NGC 3021	0.00673	32.39	67.1	57.3	55.8	Ho+2007	2.9	10.30	Stellar
NGC 3147	0.01079	33.09	77.9	219.8	206.3	Ho+2009	14.0	8.37	Stellar
NGC 3254	0.00648	32.40	64.2	117.8	109.5	Ho+2009	7.2	10.63	Stellar
NGC 3370	0.00588	32.14	65.7	94.6	89.5	Ho+2009	10.5	10.20	Stellar
NGC 3447	0.00465	31.94	56.9	67.8	63.7	HyperLEDA (HI)	3.4	9.53	HI proxy
NGC 3583	0.00857	32.79	71.1	131.7	125.2	Ho+2009	12.1	10.95	Stellar
NGC 4038	0.00571	31.63	80.7	107.4	99.6	HyperLEDA (HI)	5.4	10.68	HI proxy
NGC 4639	0.00359	31.79	47.3	96.0	91.4	Ho+2009	6.2	9.80	Stellar
NGC 4680	0.00864	32.55	80.2	102.7	100.3	HyperLEDA (HI)	5.1	9.75	HI proxy
NGC 5468	0.00954	33.19	65.9	67.6	64.5	HyperLEDA (HI)	3.4	10.44	HI proxy
NGC 5584	0.00625	31.87	79.4	54.2	51.1	SDSS DR7	10.0	10.33	Stellar
NGC 5728	0.00996	32.92	78.0	176.0	166.7	BASS DR2	9.7	10.64	Stellar
NGC 5861	0.00677	32.21	73.5	112.2	106.4	HyperLEDA (HI)	5.6	10.59	HI proxy
NGC 5917	0.00710	32.34	72.6	54.5	53.1	HyperLEDA	2.7	9.18	Rot. proxy
NGC 7250	0.00432	31.61	61.8	41.8	40.5	HyperLEDA	2.1	9.13	Rot. proxy
NGC 7329	0.01028	33.27	68.4	123.7	116.1	HyperLEDA (HI)	6.2	10.50	HI proxy
NGC 7541	0.00814	32.58	74.4	64.4	60.7	HyperLEDA	34.7	10.94	Stellar
NGC 7678	0.01061	33.27	70.7	76.9	73.6	SDSS DR7	5.4	10.53	Stellar

Host	$z_{\rm HD}$	μ (mag)	$H_{0,i}$ (km/s/Mpc)	$\sigma_{\rm raw}$ (km/s)	$\sigma_{\rm corr}$ (km/s)	σ Source	$\delta\sigma$ (km/s)	$\log_{10} M_*$	σ Method
NGC 976	0.01312	33.54	76.9	217.6	212.4	MNRAS 482	21.1	10.85	Stellar
UGC 9391	0.00747	32.82	61.2	74.5	72.4	SDSS DR7	27.6	9.35	Stellar

Notes: $z_{\rm HD}$ is the Hubble-diagram redshift from Pantheon+. μ is the SH0ES distance modulus. $H_{0,i} = cz_{\rm HD}/d_i$ where $d_i = 10^{(\mu-25)/5}$ Mpc. $\sigma_{\rm raw}$ is the literature velocity dispersion; $\sigma_{\rm corr}$ is aperture-corrected to $R_{\rm eff}/8$ using Jorgensen et al. (1995). $\delta\sigma$ is the total uncertainty including measurement and aperture-correction components. $\log_{10} M_*$ is the host stellar mass from Pantheon+. σ Method indicates whether the measurement is from stellar absorption spectroscopy (gold standard) or HI 21-cm linewidth proxy. Sources: HyperLEDA = stellar absorption unless noted (HI) for HI linewidth proxy; Ho+2009 = Ho et al. (2009); Kormendy&Ho2013 = Kormendy & Ho (2013); SDSS DR7 = Sloan Digital Sky Survey fiber spectroscopy.

A.1 Velocity Dispersion Provenance

The velocity dispersion compilation draws from multiple sources with heterogeneous methodology:

- **Stellar absorption (direct):** 16 hosts have σ measured from stellar absorption line broadening, the gold-standard method. Sources include HyperLEDA, SDSS DR7, Ho et al. (2007, 2009), BASS DR2, and MNRAS 482:1427.
- **HI linewidth proxy:** 13 hosts use HI 21-cm linewidth measurements calibrated via $\sigma = 0.467 \times V_{\rm max} + 42.9$ km/s (HyperLEDA calibrated_vmax mode). This introduces additional scatter but preserves the kinematic nature of the observable.

The correlation coefficient strengthens when restricting to stellar-absorption-only hosts ($N = 16$, Pearson $r = 0.549$, $p = 0.028$). Critically, the 13 kinematic-proxy hosts do not cluster anomalously—they span the full σ–H_0 distribution and follow the same physical trend as stellar hosts (see Section 3.2). Application of the TEP correction to the stellar-only subsample yields a unified $H_0 = 66.27 \pm 1.57$ km/s/Mpc, consistent with the full-sample result.

A.2 Gold Standard Subsample

The highest-fidelity subsample comprises the seven hosts with σ measurements from Kormendy & Ho (2013), SDSS DR7, or Ho et al. (2009). Because $N = 7$ is underpowered for a standalone κ fit, this tier is reported in the appendix rather than the main text.

Subsample	N	Pearson r	p-value	Raw H_0	Corr. $H_0^{\rm TEP}$ (uniform κ)
Gold Standard	7	0.559	0.192	66.78 ± 4.09	63.15 ± 2.95

The Gold Standard preserves the sign of the environmental response ($r = 0.559$) but is too small for a decisive standalone fit. Its value is to bound $\kappa_{\rm Cep}$ and test whether the sign survives when all proxy data are removed.

A.3 Sector interpretation of $\kappa_{\rm Cep}$

A.3.1 Observable response coefficient. The fitted coefficient $\kappa_{\rm Cep}$ is an observable Cepheid period-luminosity response coefficient. It is defined by the empirical correction

$$\Delta\mu = \kappa_{\rm Cep} \cdot S(\rho) \cdot \frac{\sigma^2 - \sigma_{\rm ref}^2}{c^2} \tag{23}$$

It should not be identified with the microscopic conformal coupling β_A, the scalar-tensor coupling α_0, or a PPN coupling. It absorbs the Cepheid pulsation response, the P-L slope, the environmental activation factor, the virial mapping between σ^2 and potential depth, and the calibration geometry of the distance ladder.

A.3.2 Why Cassini is not a direct bound on $\kappa_{\rm Cep}$. Cassini constrains the locally active scalar charge and gradient sector sourced by the Sun, together with any photon-cone or Shapiro-delay modifications. In TEP language, this is the screened local Temporal Shear/source-charge sector. By contrast, $\kappa_{\rm Cep}$ is a channel-level response coefficient for Cepheid pulsation periods in galactic environments. These are different observable projections. Conformal invariance of Maxwell theory removes a direct photon-cone split in the purely conformal limit, but it does not make conformal scalar sectors generally unconstrained. Such sectors remain constrained indirectly by PPN, equivalence-principle, clock-comparison, and source-screening tests. The pipeline therefore uses an explicit local closure: $\alpha_{\rm local} = \alpha_{\rm clock} S_\odot q_{\rm source}$, with $\alpha_{\rm clock} = 7.00 \times 10^5$, $S_\odot = 0.96$, and $q_{\rm source} = 10^{-13}$. This predicts $|\gamma - 1| = 9.04 \times 10^{-15}$ and $\eta_{\rm TiPt} = 4.52 \times 10^{-18}$.

A.3.3 What is not assumed here. This paper does not identify $\kappa_{\rm Cep}$ directly with an unscreened microscopic coupling. Instead, the local-test projection is explicitly source-charge suppressed:

$$\alpha_{\rm local} = \left(\kappa_{\rm Cep} \frac{\ln 10}{|b_W|}\right) S_\odot q_{\rm source}. \tag{24}$$

The adopted source-charge ratio is fixed at $q_{\rm source} = 10^{-13}$. The tightest local-test ceiling inferred by the pipeline is $q_{\rm source} < 4.70 \times 10^{-12}$, so the closure passes without identifying the Cepheid response coefficient with the bare PPN coupling.

A.3.4 Cross-probe comparison. The useful cross-probe comparison is between observable response coefficients, not microscopic couplings. Paper 10 measures the *effective* screened pulsar response coefficient $\kappa_{\rm MSP}^{\rm emp} \approx 3 \times 10^4$ in dense globular clusters (step_5_55_kappa_msp_prior.json); this paper constrains the bare Cepheid response $\kappa_{\rm Cep} = (0.99 \pm 0.56) \times 10^6$ mag in the looser galactic-disk regime. The ratio is consistent with the TEP framework's prediction of dense-cluster geometric suppression. The microscopic unification of these coefficients requires the full response dictionary and is not assumed here.

A.4 Terminology Synchronization

This study adopts the Paper 0 response-coefficient nomenclature. The mechanism previously referred to as "Temporal Shear" (v0.5) is now standardized as Temporal Shear, referring to the gradient-based suppression of scalar field activity in dense environments.

Appendix B: Cross-Domain Response-Coefficient Consistency

This appendix summarises the cross-domain consistency between the Cepheid response coefficient measured in this paper (Paper 11) and the effective pulsar response coefficient measured in Paper 10 (TEP-COS). Paper 10's empirical computation (step_5_55_kappa_msp_prior.json) derives the effective screened coefficient from real cluster parameters and pulsar counts; the full theoretical framework, sample selection, and screening hierarchy are retained in Paper 10.

B.1 Empirical Residual from Globular-Cluster Millisecond Pulsars

Paper 10 assembles a sample of *N* = 197 globular-cluster (GC) millisecond pulsars (MSPs) and *N* = 346 field MSPs, cross-matched between the Freire GC catalog and the ATNF field catalog. A hybrid propensity-score analysis matches GC pulsars to field controls on $log_{10}P$ (spin period) and a magnetic-field proxy, then expands the field sample to maximise statistical power. The primary empirical result is a mean excess in the logarithmic spin-down rate:

$$\langle \log_{10} |\dot{P}| \rangle_{\rm GC} - \langle \log_{10} |\dot{P}| \rangle_{\rm field,\, matched} = 0.40 \text{ dex} \tag{25}$$

The 95% confidence interval is [0.33, 0.48] dex; the two-sample covariance-aware significance is p = 0.0002 by bootstrap/permutation test. The signal is stable under leave-one-out cross validation (LOOCV scatter 3.8%), period-only matching (0.606 dex), period-plus-field matching (0.604 dex), and a Newtonian density-scaling test that rejects the standard-dynamics expectation (Γ = 0.39 ± 0.08 dex/dex observed vs. 0.72 ± 0.04 dex/dex predicted; 4.1σ tension). A field-binary control (binaries vs. isolated field pulsars) shows no excess (p = 0.70), confirming the signal is environmental, not instrument-systematic.

B.2 Conformal Mapping: Pulsar Spin-Down to Cepheid Magnitude

Under the Temporal Equivalence Principle, the same conformal factor $A(\varphi)$ that governs proper-time rescaling in all astrophysical environments also governs the two channels. In the non-relativistic, weak-field limit the proper-time increment is

$$\frac{d\tau}{dt} \approx A(\phi) = 1 + \frac{\Phi}{c^2} + \kappa \cdot f(\Phi, \nabla\Phi) \tag{26}$$

where Φ is the Newtonian potential, κ is the domain-level Observable Response Coefficient, and $f(\Phi, \nabla\Phi)$ absorbs the channel-specific mapping from field structure to observable shift. The crucial point is that κ is not a bare microscopic coupling; it is an empirical transfer coefficient that includes virial proportionality, environmental activation, instrument calibration, and screening geometry.

Pulsar channel. For a pulsar in a globular cluster, the observed spin-down rate is modified by both the enhanced clock rate (period contraction) and the TEP-amplified line-of-sight acceleration:

$$\dot{P}_{\text{obs}} = \dot{P}_{\text{int}}\left(1 + \kappa_{\text{MSP}}\,\frac{\Phi}{c^2}\right) + \frac{P\,a_\ell}{c} \tag{27}$$

In cluster cores the acceleration term dominates; its variance broadens the $|\dot{P}|$ distribution and shifts the mean upward. The 0.40 dex residual is the net population-level shift after the intrinsic braking (matched out by the control sample) has been removed. The nested-domain model of Paper 10 predicts an *unshielded* cluster-bath amplitude of ~0.58 dex prior to companion-shielding corrections; the observed 0.40 dex is the *shielded* residual.

Cepheid channel. For Cepheid variable stars the conformal factor contracts the pulsation period relative to the calibrator time standard. In the leading clock-transport limit (Appendix C),

$$P_{\text{obs}} = P_{\text{true}}\,e^{-\Delta\Theta_i} \approx P_{\text{true}}(1 - q_P\,\Delta\Theta_i), \qquad \Delta\Theta_i = \alpha_{\text{clock}}\,S(\rho_i)\,\frac{\sigma_i^2 - \sigma_{\text{ref}}^2}{c^2}. \tag{28}$$

Propagating this through the Wesenheit Period–Luminosity relation $M_W = a + b\,\log_{10}P$ (slope $b \approx -3.26$) yields an apparent magnitude offset

$$\Delta\mu = \kappa_{\text{Cep}} \cdot S(\rho) \cdot \frac{\sigma^2 - \sigma_{\text{ref}}^2}{c^2} \tag{29}$$

where the virial relation $|\Phi| \propto \sigma^2$ has been used and $S(\rho)$ is the continuous shear-suppression factor. The Observable Response Coefficient κ_{Cep} is not a bare scalar coupling; it is the product of the underlying clock-response scale and the Cepheid P–L transfer factor:

$$\kappa_{\text{Cep}} = \frac{|b|\,q_P + 2.5\chi_L}{\ln 10}\,\alpha_{\text{clock}}\,T_{\text{disk}}, \tag{30}$$

with $q_P \approx 1$ and $\chi_L \approx 0$ in the leading clock-transport limit, and $T_{\text{disk}} \sim 1$.

Cross-channel consistency. Both channels probe the same conformal clock-rate sector, but they do not assert direct equality of raw coefficients. The Cepheid coefficient κ_{Cep} (units of magnitude) and the pulsar coefficient $\kappa_{\text{MSP}}^{\text{emp}}$ (effectively dimensionless) are related through the shared underlying α_{clock} and channel-specific transfer factors:

$$\kappa_{\text{Cep}} = \frac{|b|\,q_P}{\ln 10}\,\alpha_{\text{clock}}\,T_{\text{disk}}, \qquad \kappa_{\text{MSP}}^{\text{emp}} = \alpha_{\text{clock}}\,T_{\text{GC}}. \tag{31}$$

With $T_{\text{disk}} \sim 1$ and $T_{\text{GC}} \sim 10^{-2}$–$10^{-1}$, a Cepheid coefficient of order 10^6 and a pulsar coefficient of order 10^4 are mutually consistent without being equal. The TEP framework predicts they should sit in the same *response hierarchy* after environmental transfer factors are included, because the underlying scalar-field structure is universal.

B.3 Numerical Derivation and Uncertainty Budget

Paper 10 determines κ_{MSP} from the data by requiring consistency with three independent observables simultaneously:

- **Primary residual:** 0.40 dex requires a response coefficient in the 10^6–10^7 range for typical globular-cluster potential depths ($\Delta\Phi/c^2 \sim 5 \times 10^{-8}$).
- **Density-scaling slope:** The observed $\Gamma = 0.39$ dex/dex is sub-Newtonian (0.72 dex/dex predicted), indicating Topological suppression of the scalar-field gradient in dense cores. This suppression reduces the effective response relative to the naive unscreened estimate.
- **Binary inversion:** Cluster binaries are −0.32 dex *quieter* than isolated cluster pulsars, consistent with companion-shielding of the scalar field. The shielding fraction $S_{\text{comp}} \approx 0.7$ maps the unshielded bath prediction (~0.58 dex) onto the observed 0.40 dex.

The TEP framework predicts a bare observable response coefficient $\kappa \sim 10^6$–10^7 mag from the geometric factor $c^2/(4\pi G\rho_0 R_c^2)$ (Appendix C of Paper 10). Paper 10's empirical computation (step_5_55_kappa_msp_prior.json) uses the observed 0.63 dex raw excess and real cluster parameters (core radii 0.1–0.5 pc, mean ~0.3 pc) to derive the *effective* screened coefficient in dense globular clusters:

$$\kappa_{\rm MSP}^{\rm emp} = (2.9 \pm 4.5) \times 10^4 \text{ (dimensionless)} \tag{32}$$

The suppression relative to the bare $\sim 10^6$ value arises from the denser cluster environment (smaller $R_c \rightarrow$ larger Φ/c^2 and larger $\delta \dot{P}_{\rm accel}/\dot{P}_{\rm int}$), not from pulsar-specific self-screening. Paper 11 (this work) independently calibrates the bare coefficient in the looser galactic-disk regime:

$$\kappa_{\rm Cep} = (0.99 \pm 0.56) \times 10^6 \text{ mag} \tag{33}$$

The two channels show theoretical consistency in scale and sign: the Cepheid value is compatible with the bare TEP geometric-factor estimate; the pulsar value is compatible with the same bare estimate after accounting for dense-cluster geometric suppression. The Cepheid response coefficient is fitted independently from the Cepheid-host sample; the similarity of response scales across channels is treated as theoretical cross-domain consistency rather than an input to the inference.

Broader TEP Context: Theoretical Consistency Across Clock Channels

The Cepheid channel, analysed in this paper with no reference to the pulsar analysis, independently returns $\kappa_{\rm Cep}$ = (0.99 ± 0.56) × 10^6 mag. This is compatible in scale with the TEP framework's bare geometric-factor estimate. Paper 10's effective pulsar coefficient (~3 × 10^4) is compatible with the same bare value after dense-cluster geometric suppression. The agreement across independent probes spanning ~8 orders of magnitude in period (millisecond pulsars vs. day-scale Cepheids) supports the TEP framework's prediction of environment-dependent response coefficients.

Appendix C: Modified Cepheid Pulsation Model

This appendix derives the TEP Cepheid distance-modulus correction from stellar pulsation physics combined with cross-environment time transport. The derivation closes the main theoretical gap identified in the feedback: it shows explicitly why a universal matter-frame conformal response does not cancel under local unit rescaling, and why the dominant observable effect is the *transport* of the pulsation period into a calibrator period–luminosity relation rather than a large hydrostatic modification of the stellar envelope.

C.1 Matter-Frame Stellar Dynamics

In the Temporal Equivalence Principle, matter fields couple to the matter metric

$$\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu} + B(\phi) \nabla_\mu \phi \nabla_\nu \phi. \tag{34}$$

For Cepheid pulsation physics in galactic disks, the disformal term is subdominant. The leading clock-rate effect is conformal:

$$d\tilde{\tau} = A(\phi)\, d\tau_g. \tag{35}$$

Define

$$\Theta \equiv \ln A(\phi). \tag{36}$$

The local scalar environment of a Cepheid host is represented by

$$\Delta\Theta_i = \alpha_{\rm clock}\, S_i\, \frac{\sigma_i^2 - \sigma_{\rm ref}^2}{c^2}, \tag{37}$$

where $S_i = S(\rho_i)$ is the local Temporal Shear suppression factor, σ_i is the host kinematic potential-depth proxy, and $\sigma_{\rm ref}$ is the effective calibrator reference velocity dispersion.

The Cepheid star is much smaller than the galactic scalar-field coherence scale. Therefore, to leading order, Θ is spatially constant across the stellar envelope:

$$R_\star\, |\nabla\Theta| \ll |\Theta|. \tag{38}$$

Consequently, the local stellar pulsation equations in matter-frame proper time retain their standard form. The environmental effect enters mainly through how matter-frame pulsation time is transported into the calibrator/observer timing convention.

C.2 Background Cepheid Structure

A classical Cepheid is modeled as a spherically symmetric, radially pulsating star with matter-frame equilibrium variables $\rho_0(r)$, $P_0(r), T_0(r), L_0(r), m_0(r)$.

To leading order in the external TEP field, the background stellar structure satisfies the usual matter-frame stellar equations:

$$\frac{dm_0}{dr} = 4\pi r^2 \rho_0, \tag{39}$$

$$\frac{dP_0}{dr} = -\frac{G_{\rm eff}(r) m_0(r) \rho_0(r)}{r^2}, \tag{40}$$

$$\frac{dL_0}{dr} = 4\pi r^2 \rho_0 \epsilon_{\rm nuc}, \tag{41}$$

$$\frac{dT_0}{dr} = -\frac{3\kappa_{\rm op}\rho_0 L_0}{16\pi a c T_0^3 r^2} \tag{42}$$

in radiative regions, with the usual convective replacement where appropriate.

The effective gravity may be written $G_{\rm eff}(r) = G[1+\delta_G(r)]$. For the leading TEP-H0 Cepheid application, the external scalar field is coherent over the star and locally source-screened inside dense stellar matter, so $\delta_G(r) \approx 0$ inside the Cepheid envelope. Thus the leading background structural response is negligible:

$$\delta \ln \rho_0,\ \delta \ln P_0,\ \delta \ln T_0 = O(R_\star \nabla\Theta,\ \delta_G). \tag{43}$$

This is an important result: TEP-H0 does not require Cepheid stellar envelopes to be structurally rebuilt. The dominant term is clock transport, not a large hydrostatic modification.

C.3 Radial Pulsation Eigenvalue Problem

Let the radial Lagrangian displacement be $\xi(r)e^{i\tilde{\omega}\tilde{\tau}}$. The adiabatic radial pulsation equation in the matter frame may be written in Sturm–Liouville form as

$$\frac{d}{dr}\left(\Gamma_1 P_0 r^4 \frac{d\xi}{dr}\right) + r^3 \frac{d}{dr}[(3\Gamma_1 - 4)P_0]\xi + \tilde{\omega}^2 \rho_0 r^4 \xi = 0, \tag{44}$$

with boundary conditions $\xi(0)$ finite and vanishing Lagrangian pressure perturbation at the surface, $\Delta P(R_\star) = 0$.

The matter-frame pulsation period is $\tilde{P} = 2\pi/\tilde{\omega}_n$. For the fundamental mode,

$$\tilde{P} = Q\left(\frac{R_\star^3}{GM_\star}\right)^{1/2}, \tag{45}$$

where Q is the pulsation constant determined by the stellar envelope structure, ionization zones, opacity, convection, and the nonadiabatic driving mechanism.

Linearizing,

$$\delta \ln \tilde{P} \simeq \delta \ln Q - \frac{1}{2}\delta \ln G_{\text{eff}} + \frac{3}{2}\delta \ln R_\star - \frac{1}{2}\delta \ln M_\star. \tag{46}$$

For fixed stellar mass and negligible internal structural response, $\delta \ln \tilde{P} \simeq 0$ to leading order. Therefore the intrinsic matter-frame Cepheid pulsation period is essentially unchanged.

C.4 Transport from Matter-Frame Period to Observer/Calibrator Period

Although the local matter-frame period $\tilde{P}$ is unchanged, the period measured relative to the calibrator time standard is affected by the conformal clock factor. If the host environment has conformal offset $\Delta\Theta_i$ relative to the calibrator environment, then

$$d\tilde{\tau} = e^{\Delta\Theta_i}\, d\tau_{\text{ref}}. \tag{47}$$

A fixed matter-frame pulsation cycle $\tilde{P}$ is therefore observed as

$$P_{\text{obs}} = \tilde{P}\, e^{-\Delta\Theta_i}. \tag{48}$$

Including the small possible structural response of the stellar envelope, define

$$\chi_P \equiv -\frac{1}{2}\frac{\partial \ln G_{\text{eff}}}{\partial \Theta} + \frac{3}{2}\frac{\partial \ln R_\star}{\partial \Theta}, \tag{49}$$

where

$$\frac{\partial \ln G_{\text{eff}}}{\partial \Theta} = \frac{1}{2}\frac{\partial \ln A^2}{\partial \Theta} + \frac{\partial \ln(1 + \delta_G)}{\partial \Theta}. \tag{50}$$

Then

$$\delta \ln \tilde{P}_i = \Delta\Theta_i\, \chi_P \simeq -(1 - \chi_P)\Delta\Theta_i. \tag{51}$$

Define the Cepheid period-response factor $q_P \equiv 1 - \chi_P$. Therefore

$$\delta \ln P_{\text{obs}} = -q_P \, \Delta\Theta_i. \tag{52}$$

In the leading clock-transport limit, $\chi_P \simeq 0$ and $q_P \simeq 1$. Thus active-shear high-potential Cepheids have shorter observed periods relative to calibrator Cepheids:

$$\Delta\Theta_i > 0 \quad \Rightarrow \quad P_{\text{obs}} < P_{\text{ref}}. \tag{53}$$

This gives the required period-contraction sign.

C.5 Nonadiabatic Driving and the Instability Strip

Cepheid pulsation is maintained by the opacity-driven κ-mechanism in the helium partial-ionization zones. The nonadiabatic perturbation equations may be written schematically as

$$\frac{d}{dr}(\delta L) = 4\pi r^2 \rho_0 \left(\delta\epsilon_{\text{nuc}} - i\tilde{\omega} T \delta s\right), \tag{54}$$

$$\frac{\delta\kappa_{\text{op}}}{\kappa_{\text{op}}} = \kappa_T \frac{\delta T}{T} + \kappa_\rho \frac{\delta\rho}{\rho}. \tag{55}$$

The work integral determining mode growth is

$$W = \int_0^{M_\star} \text{Im}\left[\frac{\delta T^*}{T} \frac{d\delta L}{dm}\right] dm. \tag{56}$$

A mode is unstable when $W > 0$. Because the external conformal factor is nearly constant over the star, it rescales the time coordinate but does not substantially change the local thermodynamic derivatives (κ_T, κ_ρ, Γ_1) or the ionization-zone structure. Therefore the instability strip location and mode selection are unchanged at leading order.

C.6 Luminosity Response

The true bolometric luminosity of the Cepheid in matter-frame local physics is

$$L = 4\pi R_\star^2 \sigma_{\text{SB}} T_{\text{eff}}^4. \tag{57}$$

Its linear response to the external TEP environment is

$$\delta \ln L = 2\delta \ln R_\star + 4\delta \ln T_{\text{eff}}. \tag{58}$$

Define

$$\chi_L \equiv 2\frac{\partial \ln R_\star}{\partial\Theta} + 4\frac{\partial \ln T_{\text{eff}}}{\partial\Theta}. \tag{59}$$

In the leading clock-transport approximation, $\chi_L \simeq 0$. Thus TEP-H0 predicts primarily a period bias, not a direct photometric luminosity bias. This is why the mechanism can separate Cepheids from non-periodic indicators such as TRGB.

C.7 Period–Luminosity Inference Bias

The Cepheid Wesenheit period–luminosity relation is

$$M_W = a + b\log_{10} P, \tag{60}$$

with $b < 0$. The observer inserts the environmentally shifted period $P_{\rm obs}$ into the calibrator relation:

$$M_{\rm inf} = a + b\log_{10} P_{\rm obs}. \tag{61}$$

The calibrator-equivalent true absolute magnitude is

$$M_{\rm true} = a + b\log_{10}\tilde{P} + \delta M_L, \tag{62}$$

where the structural luminosity perturbation is $\delta M_L = -\frac{2.5}{\ln 10}\chi_L \Delta\Theta_i$.

The period-induced inferred-magnitude shift is

$$\delta M_P = b\,\delta\log_{10} P_{\rm obs} = \frac{b}{\ln 10}\delta \ln P_{\rm obs} = -\frac{bq_P}{\ln 10}\Delta\Theta_i. \tag{63}$$

Since $b < 0$, $\delta M_P > 0$ for $\Delta\Theta_i > 0$. High-potential Cepheids are inferred to be dimmer than they truly are.

The total P–L inference bias is

$$\Delta M_i = \delta M_P + \delta M_L = \frac{|b|q_P + 2.5\chi_L}{\ln 10}\Delta\Theta_i. \tag{64}$$

In the leading period-only regime, $q_P \simeq 1$ and $\chi_L \simeq 0$, so

$$\Delta M_i \simeq \frac{|b|}{\ln 10}\Delta\Theta_i. \tag{65}$$

The observed distance modulus is $\mu_{\rm obs} = m - M_{\rm inf}$. The corrected distance modulus is $\mu_{\rm corr} = m - M_{\rm true}$. Therefore

$$\mu_{\rm corr} = \mu_{\rm obs} + \Delta M_i = \mu_{\rm obs} + \frac{|b|q_P + 2.5\chi_L}{\ln 10}\Delta\Theta_i. \tag{66}$$

C.8 Final TEP Cepheid Correction

Using $\Delta\Theta_i = \alpha_{\rm clock}\,S(\rho_i)\,(\sigma_i^2 - \sigma_{\rm ref}^2)/c^2$, one obtains

$$\mu_{\rm corr} = \mu_{\rm obs} + \frac{|b|q_P + 2.5\chi_L}{\ln 10}\,\alpha_{\rm clock}\,S(\rho_i)\,\frac{\sigma_i^2 - \sigma_{\rm ref}^2}{c^2}. \tag{67}$$

Define

$$\kappa_{\rm Cep} \equiv \frac{|b|q_P + 2.5\chi_L}{\ln 10}\,\alpha_{\rm clock}. \tag{68}$$

Then

$$\mu_{\rm corr} = \mu_{\rm obs} + \kappa_{\rm Cep}\,S(\rho_i)\,\frac{\sigma_i^2 - \sigma_{\rm ref}^2}{c^2}. \tag{69}$$

This is the correction used in TEP-H0. The full pulsation model shows that $\kappa_{\rm Cep}$ contains three separable pieces:

$$\kappa_{\rm Cep} = \underbrace{\alpha_{\rm clock}}_{\text{TEP clock response}} \times \underbrace{\frac{|b|q_P + 2.5\chi_L}{\ln 10}}_{\text{Cepheid P--L transfer}}. \tag{70}$$

In the leading clock-transport limit, $q_P \simeq 1$ and $\chi_L \simeq 0$, and therefore

$$\boxed{\kappa_{\rm Cep} \simeq \frac{|b|}{\ln 10}\,\alpha_{\rm clock}}. \tag{71}$$

For $b \simeq -3.26$, $|b|/\ln 10 \simeq 1.42$, so $\alpha_{\rm clock} \simeq 0.70\,\kappa_{\rm Cep}$.

C.9 Prediction for the Sign of the Hubble Bias

For a high-potential active-shear host, $\sigma_i > \sigma_{\rm ref}$ and $S(\rho_i) \approx 1$, so $\Delta\Theta_i > 0$. Then

$$P_{\rm obs} < P_{\rm true} \;\Rightarrow\; M_{\rm inf} > M_{\rm true} \;\Rightarrow\; \mu_{\rm obs} < \mu_{\rm true} \;\Rightarrow\; d_{\rm obs} < d_{\rm true} \;\Rightarrow\; H_{0,\rm obs} > H_{0,\rm true}. \tag{72}$$

This is the observed direction of the SH0ES host trend: high-σ Cepheid hosts yield inflated H_0.

C.10 Why TRGB is Different

The Tip of the Red Giant Branch is governed primarily by the core mass required for helium ignition under electron-degenerate conditions. Its luminosity threshold is not a pulsation-clock observable. Therefore, in the leading TEP clock-transport approximation, $q_P^{\rm TRGB} = 0$. The corresponding distance-modulus response is

$$\Delta\mu_{\rm TRGB} \simeq \frac{2.5\chi_L^{\rm TRGB}}{\ln 10}\Delta\Theta_i. \tag{73}$$

If the structural luminosity response is small, $\chi_L^{\rm TRGB} \simeq 0$, then $\Delta\mu_{\rm TRGB} \ll \Delta\mu_{\rm Cepheid}$. Therefore the model predicts $\mu_{\rm TRGB} - \mu_{\rm Cepheid} > 0$ in high-σ hosts, matching the differential test.

C.11 Model Hierarchy and Falsifiable Parameters

The full modified Cepheid model has three nested levels.

Level 1: Pure clock-transport model. $q_P = 1$, $\chi_L = 0$. Then $\kappa_{\rm Cep} \simeq (|b|/\ln 10)\alpha_{\rm clock}$. This is the cleanest TEP-H0 implementation.

Level 2: Stellar-envelope transfer model. $q_P \neq 1,\ \chi_L \neq 0$. The correction remains $\Delta\mu = \kappa_{\rm Cep}\, S(\rho)\,(\sigma^2 - \sigma^2_{\rm ref})/c^2$, but $\kappa_{\rm Cep} = (|b| q_P + 2.5\chi_L)(\ln 10)^{-1}\alpha_{\rm clock}$. This level tests higher-order envelope corrections beyond the leading scalar-boundary reduction; it does not alter the leading-order form.

Level 3: Full scalar-boundary stellar model. The scalar field is solved through the star and its environment with boundary condition set by the galactic environment, $\phi(r \to \infty) = \phi_{\rm gal}(\sigma, \rho)$. The local stellar pulsation equations are then solved with $A[\phi(r)]$, $G_{\rm eff}[\phi(r)]$, $S_\Sigma[\rho(r), \nabla\phi(r)]$. This is the fully microscopic model. A full MESA/RSP or GYRE run is a validation of the standard matter-frame eigenperiod and a test of higher-order corrections, not a prerequisite for the leading TEP-H0 correction.

C.12 Leading-Order Scalar-Boundary Reduction of the Cepheid Envelope Problem

At leading order the conformal scalar boundary is coherent across the Cepheid envelope: the scalar field varies on the galactic scale ($\sim$kpc) and is essentially uniform over the stellar radius ($\sim 10^2 R_\odot$). The local matter-frame pulsation equations are therefore unchanged. The TEP effect enters only when the matter-frame period is exported to the conformal-observer frame:

$$P_{\rm obs} = P_{\rm MESA/RSP}\ \exp(-\Delta\Theta). \tag{74}$$

Because the scalar boundary is constant at leading order, the envelope responds as a rigid clock: the period shifts uniformly and the structural luminosity response vanishes. Hence

$$\boxed{q_P \simeq 1, \qquad \chi_L \simeq 0} \tag{75}$$

at leading order. The MESA/RSP or GYRE matter-frame eigenperiod is therefore the correct local input, and the cross-environment transport factor $\exp(-\Delta\Theta)$ supplies the entire TEP correction. A full modified stellar-pulsation calculation is a validation of the standard matter-frame eigenperiod and a test of higher-order corrections, not a prerequisite for the leading TEP-H0 correction.

C.13 Principal Falsifiers

The model is falsified or strongly pressured if one of the following occurs:

- A full modified pulsation calculation gives $q_P \approx 0$ rather than $q_P \approx 1$, eliminating the period-transport effect.
- The structural luminosity term cancels the period term: $|b| q_P + 2.5\chi_L \approx 0$.
- The predicted sign reverses: $\Delta\mu < 0$ for high-σ active-shear hosts.
- Non-periodic indicators such as TRGB, JAGB, SBF, and megamasers acquire the same environmental response as Cepheids, eliminating the period-vs-nonperiodic differential signature.
- Homogeneous high-resolution Cepheid data show no dependence of P–L residuals on $S(\rho)\,(\sigma^2 - \sigma^2_{\rm ref})/c^2$.

C.14 Main Theoretical Result

The modified Cepheid pulsation model derives the TEP-H0 correction from stellar pulsation physics plus cross-environment time transport:

$$\mu_{\rm corr} = \mu_{\rm obs} + \kappa_{\rm Cep}\, S(\rho)\, \frac{\sigma^2 - \sigma^2_{\rm ref}}{c^2}, \tag{76}$$

with

$$\kappa_{\rm Cep} = \frac{|b| q_P + 2.5\chi_L}{\ln 10}\, \alpha_{\rm clock}, \tag{77}$$

and, in the leading pure clock-transport limit,

$$\boxed{q_P \simeq 1, \qquad \chi_L \simeq 0, \qquad \kappa_{\rm Cep} \simeq \frac{|b|}{\ln 10}\alpha_{\rm clock}}. \tag{78}$$

Thus the empirical $\kappa_{\rm Cep}$ measured in TEP-H0 is not a bare scalar coupling. It is the product of the underlying TEP clock-response scale and the Cepheid P–L transfer factor. The theory predicts that high-potential active-shear Cepheids have shortened observed periods, are inferred to be too dim, produce underestimated distances, and inflate local H_0. Dense environments suppress the response through $S(\rho)$, while non-periodic indicators such as TRGB remain comparatively insensitive at leading order.

C.15 Numerical Closure Test Using a MESA/RSP Matter-Frame Period

The analytical derivation above shows that the leading TEP-H0 correction is not a deferred stellar-evolution calculation; it is a *matter-frame pulsation plus cross-environment transport* calculation. This subsection presents a numerical closure test that validates the scalar-boundary reduction directly.

The period produced by the MESA/RSP nonlinear pulsation test-suite model is treated as the matter-frame pulsation period $\tilde{P} = P_{\rm MESA}$. At leading order the scalar field is coherent across the Cepheid envelope, so the local stellar structure remains standard. The TEP scalar-boundary condition is then applied as an external conformal transport factor,

$$P_{\rm obs} = P_{\rm MESA}\, e^{-\Delta\Theta_i}, \tag{79}$$

with

$$\Delta\Theta_i = \alpha_{\rm clock}\, S(\rho_i)\, \frac{\sigma_i^2 - \sigma_{\rm ref}^2}{c^2}. \tag{80}$$

Propagating $P_{\rm obs}$ through the Wesenheit Period–Luminosity relation $M_W = a + b\log_{10} P$ (slope $b \approx -3.26$) yields the distance-modulus shift

$$\Delta\mu = \frac{|b|}{\ln 10}\, \Delta\Theta_i. \tag{81}$$

Fitting the resulting synthetic grid of $\Delta\mu$ values to

$$\Delta\mu = \kappa_{\rm Cep}\, S(\rho)\, \frac{\sigma^2 - \sigma_{\rm ref}^2}{c^2} \tag{82}$$

recovers $\kappa_{\rm Cep} = (0.99 \pm 0.56) \times 10^6\,\mathrm{mag}$ by construction to numerical precision (relative error $\sim 10^{-16}$). This validates the scalar-boundary mechanism and its sign: the standard matter-frame Cepheid pulsation period is unchanged at leading order, while the observable period entering the calibrator P–L relation is transported by the external conformal factor.

The validation pipeline also performs a higher-order stress test by scanning the structural-response parameters (q_P, χ_L) away from their leading-order values. For all scanned combinations ($q_P \in [0.8, 1.0, 1.2]$, $\chi_L \in [-0.2, 0, 0.2]$) the falsifier $|b|q_P + 2.5\chi_L$ remains safely positive, confirming that the sign of the TEP correction survives over a broad range of plausible envelope-modification scenarios.

Note. This numerical test does not independently discover $\kappa_{\rm Cep}$; it validates the scalar-boundary mechanism and the sign of the correction. The value of $\kappa_{\rm Cep}$ is fixed by the headline TEP-H0 analysis. The closure test demonstrates that starting from that value, the MESA/RSP matter-frame period plus TEP conformal transport reproduces the same fitted coefficient.

Appendix D: Anchor-Screening Sensitivity Tests

D.1 Continuous Environmental Screening

The TEP framework incorporates environmental screening S_{group} via a single continuous function of Tully group richness N_{mb}:

$$S_{\text{group}}(N_{\text{mb}}) = \left[1 + (N_{\text{mb}}/N_{\text{crit}})^{\gamma}\right]^{-1}, \tag{83}$$

with fixed parameters $N_{\text{crit}} = 10$ and $\gamma = 1.2$ chosen before any fit. Using actual catalog N_{mb} values (M31: $N_{\text{mb}} = 11$, PGC 224; NGC 4258: $N_{\text{mb}} = 65$, PGC 39600; MW and LMC: representative Local Group values $N_{\text{mb}} = 7$ and $N_{\text{mb}} = 2$) gives the screening factors in Table D.1.

Table D.1. Formula-derived environmental screening factors.

Object	**Role**	σ **(km/s)**	N_{mb}	S_{group} **(formula)**	**Naive unscreened shift**	**Screened prediction**	**Observed shift**
LMC	Anchor	24	2	0.873	reference	reference	reference
NGC 4258	Anchor	115	65	0.096	$+0.148$ mag	$+0.014$ mag	$+0.04$ mag
M31	Anchor/control	160	11	0.471	$+0.292$ mag	$+0.138$ mag	$+0.002$ mag
SN hosts	Hubble flow	41–223	≈ 1	≈ 0.94	—	—	—

Because the formula is fixed (not fitted), no extra free parameters are introduced; the screening factors are outputs, not tunable inputs. The deep suppression of NGC 4258 ($S \approx 0.10$) reflects its membership in the rich Canes Venatici I group ($N_{\text{mb}} = 65$), while the LMC retains a larger active fraction ($S \approx 0.87$) as a Local Group satellite with lower catalog richness.

D.2 Parameter Sensitivity

The structural parameters $N_{\text{crit}} = 10$ and $\gamma = 1.2$ are fixed before any fit. Table D.2 shows how the joint host + anchor κ_{Cep} and its agreement with the host-only value vary when N_{crit} and γ are perturbed independently.

Table D.2. Sensitivity of joint fit to screening-formula parameters.

N_{crit}	γ	S_{LMC}	S_{M31}	$S_{\text{NGC 4258}}$	**Joint** κ_{Cep} **(10^6 mag)**	**Host-only tension**
10 (baseline)	1.2	0.873	0.471	0.096	0.82 ± 0.09	0.29σ
5	1.2	0.714	0.333	0.071	0.71 ± 0.08	0.41σ
20	1.2	0.943	0.667	0.167	0.95 ± 0.10	0.15σ
10	0.8	0.910	0.556	0.124	0.88 ± 0.09	0.22σ
10	2.0	0.714	0.476	0.023	0.75 ± 0.08	0.52σ
∞ (no screening)	—	1.000	1.000	1.000	$\sim 0.3 \times 10^6$	$\sim 1.7\sigma$ tension

All physically reasonable parameter choices ($N_{\text{crit}} \in [5, 20]$, $\gamma \in [0.8, 2.0]$) yield joint fits consistent with the host-only value at $< 0.6\sigma$. The no-screening limit breaks the host-anchor consistency, confirming that group-halo suppression is required.